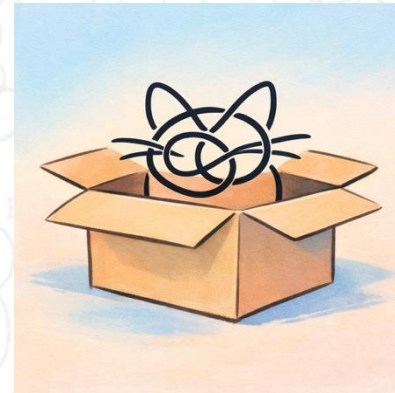


Quantum Algorithms for Link Invariants

Konstantinos Meichanetzidis
Quantum Applications, Algorithms, AI

Quantinuum



2503.05625
2210.11127
2103.06914
1807.02119
1804.01097

Coauthors on this line of work:

Tuomas Laakkonen, Enrico Rinaldi, Chris N. Self, Eli Chertkov, Matthew DeCross, David Hayes, Brian Neyenhuis, Marcello Benedetti, Sofyan Iblisdir, Gavin K. Brennen, Alex Townsend-Teague, Stefanos Kourtis, Zhi-Cheng Yang, Claudio Chamon

KNOTS

and links

Real life:

Keeping records, tying shoes, tying ties, rock climbing,
sailing, tangled earphones,
DNA, proteins, macromolecules ...

Physics:

Lord Kelvin's idea for periodic table:
elements are different knots in the Ether.
Vortex lines (fluids, plasma)

Theoretical Physics:

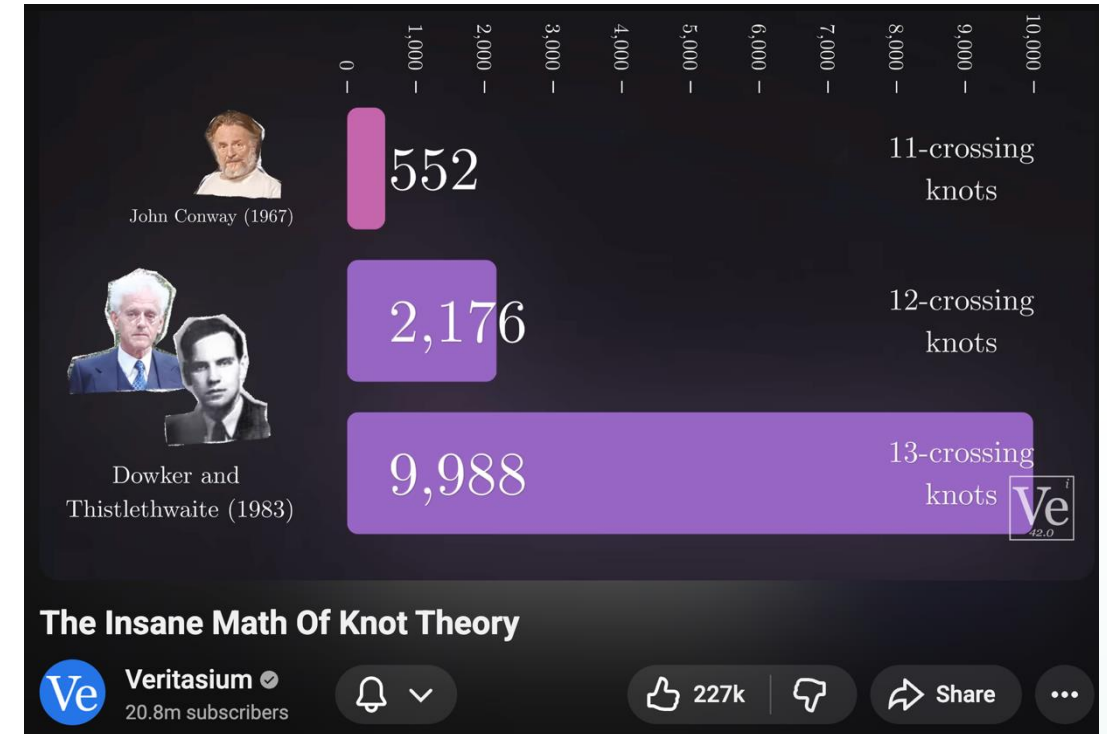
Statistical mechanics (partition functions of Potts or vertex models).
Condensed matter physics;
classification of phases of matter (topological invariants).
Witten connection with Chern Simons TQFT.

Computation:

TQC; links as quantum evolutions via braiding of anyons.
Cryptography, quantum money.
QEC with topological codes.
Knot invariants exp quad advantage.

Foundations:

Kauffman's Fibonacci universe, relational time, Helon model, LQG, Strings, ...

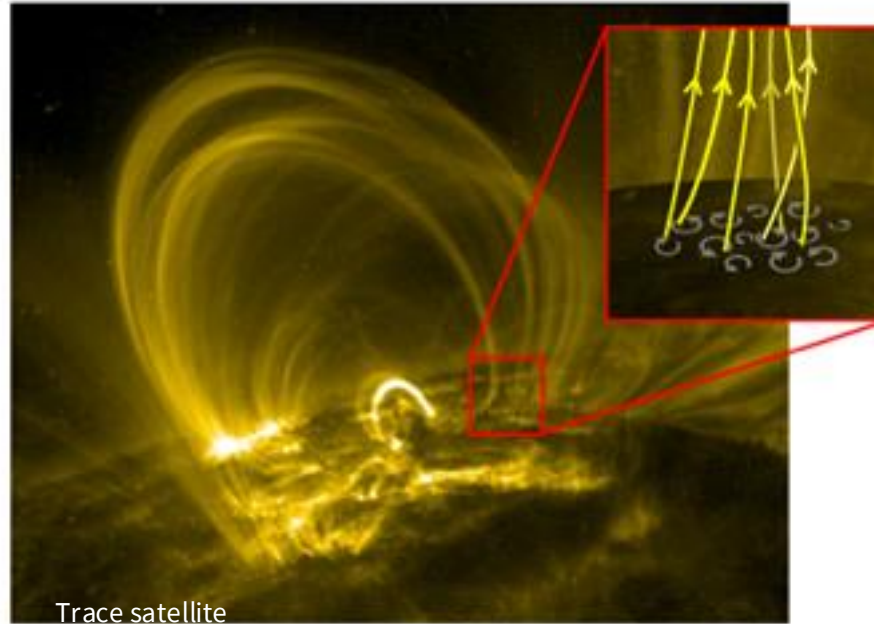
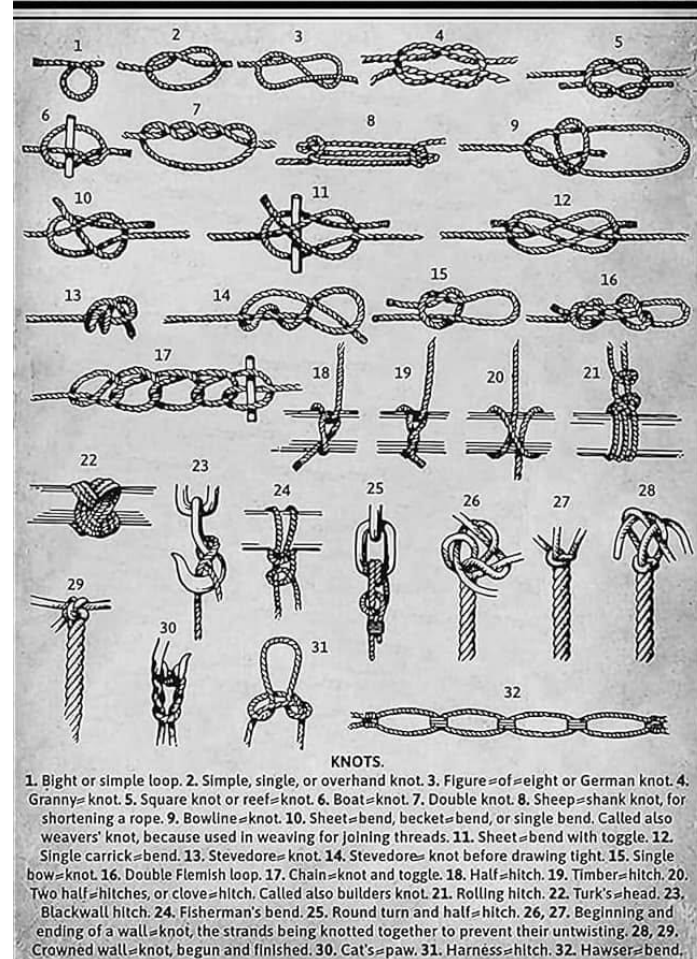


Really good Veritasium video on Knots

https://youtu.be/8DBhTXM_Br4?si=aq07miTlCxpjz457

Keeping records, tying shoes, tying ties, rock climbing, sailing, tangled earphones, DNA, proteins, macromolecules (recombination, gene expression) ...

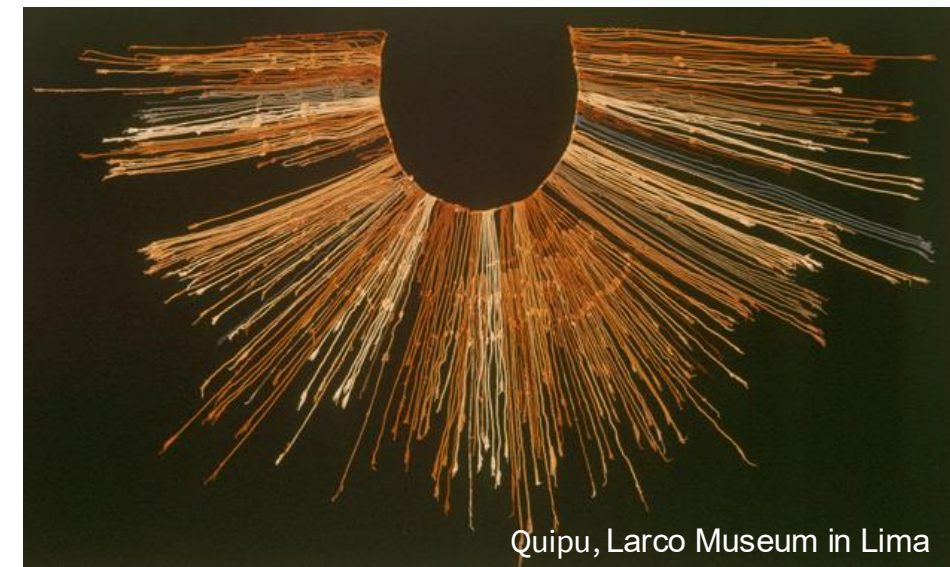
TYPES OF SAILING KNOTS



Jean-Simon Berthélemy: Alexander cuts the Gordian Knot



Science **229**, 171 (1985); copyright AAAS



Quipu, Larco Museum in Lima

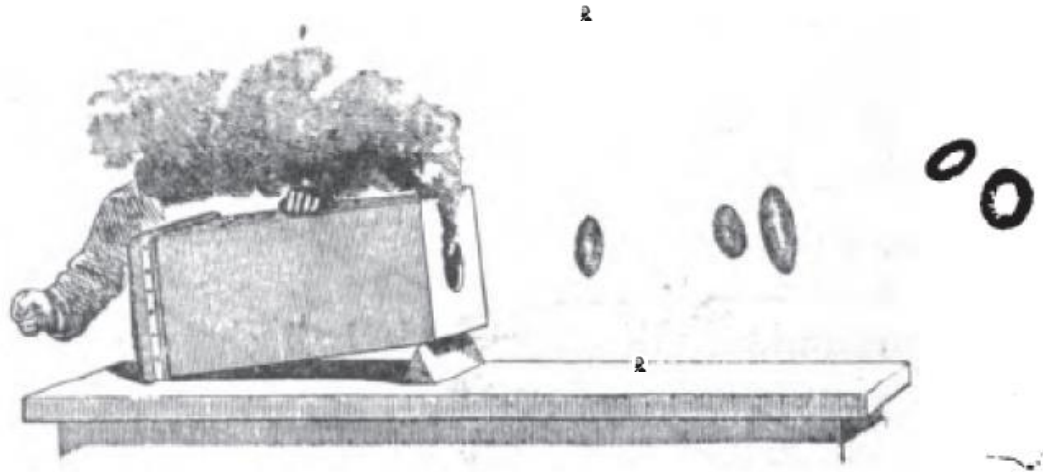
A beautiful but sadly falsified atomic theory

string theory before it was cool

Da Vinci study of water vortices



Peter Tait's vortex ring gun

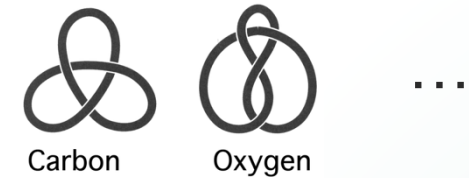


William Thomson (Lord Kelvin) 1867 :

Atoms: stable, knotted closed loops of localized vorticity (vortex rings) within the ether.

Atom stability: guaranteed by the topological permanence of the vortex in an ideal fluid.

Different chemical elements correspond to different knot types. Unknot is hydrogen, and heavier elements more complex knots.

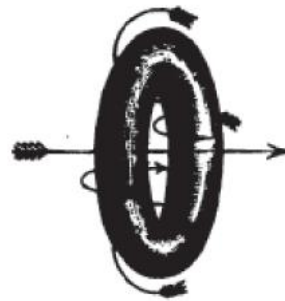


Hermann von Helmholtz:

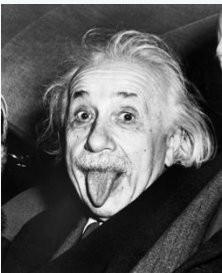
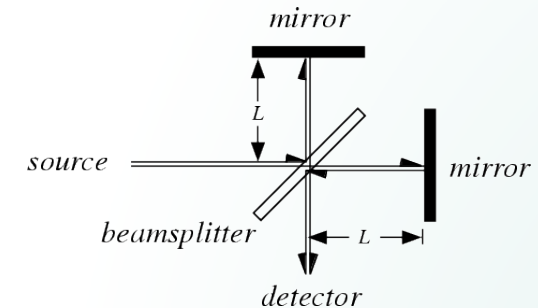
The strength of a vortex line is constant along its length.

A vortex line cannot end in a fluid; it must extend to the boundaries of the fluid or form a closed path.

A fluid element that is initially irrotational remains irrotational.

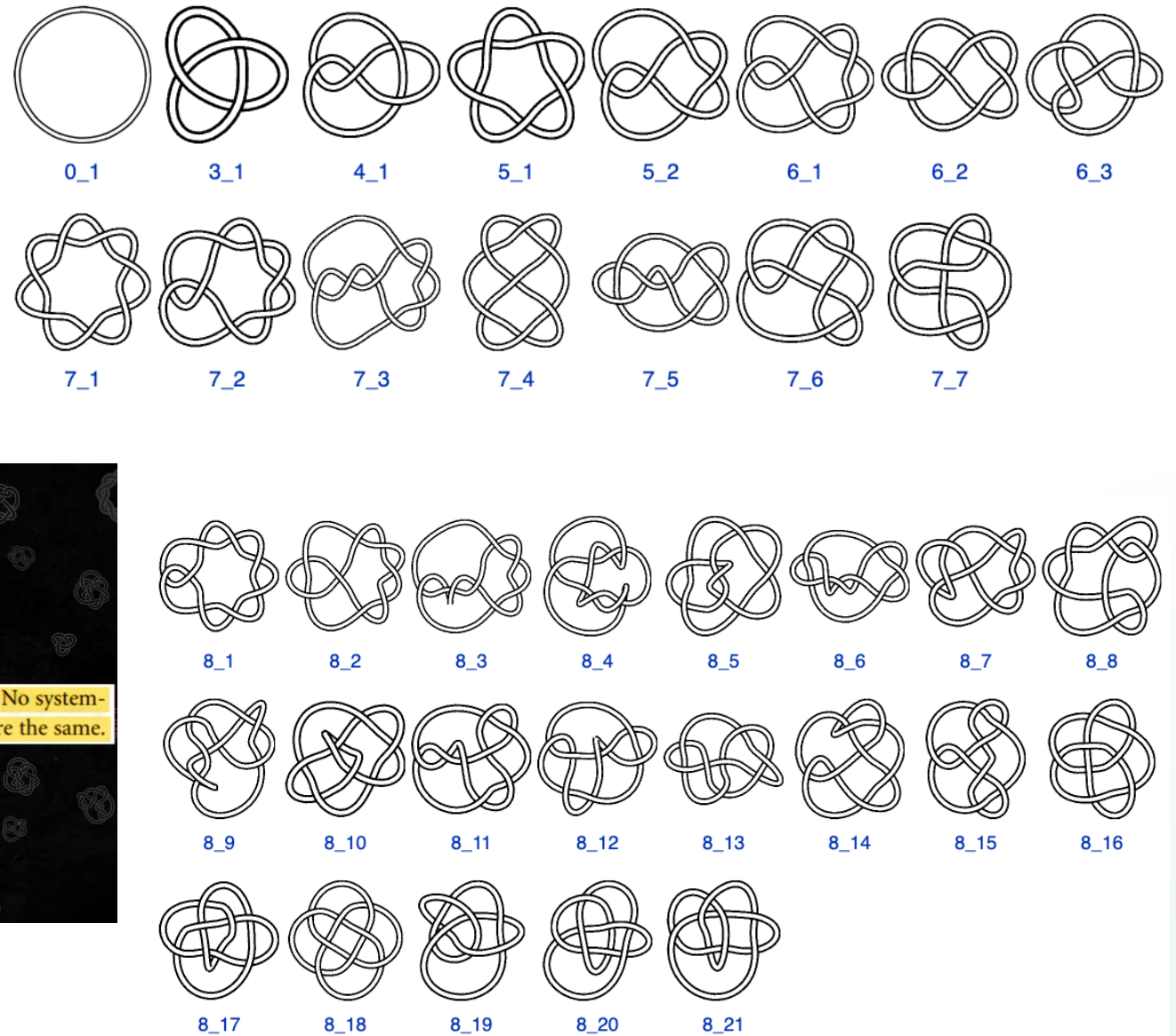
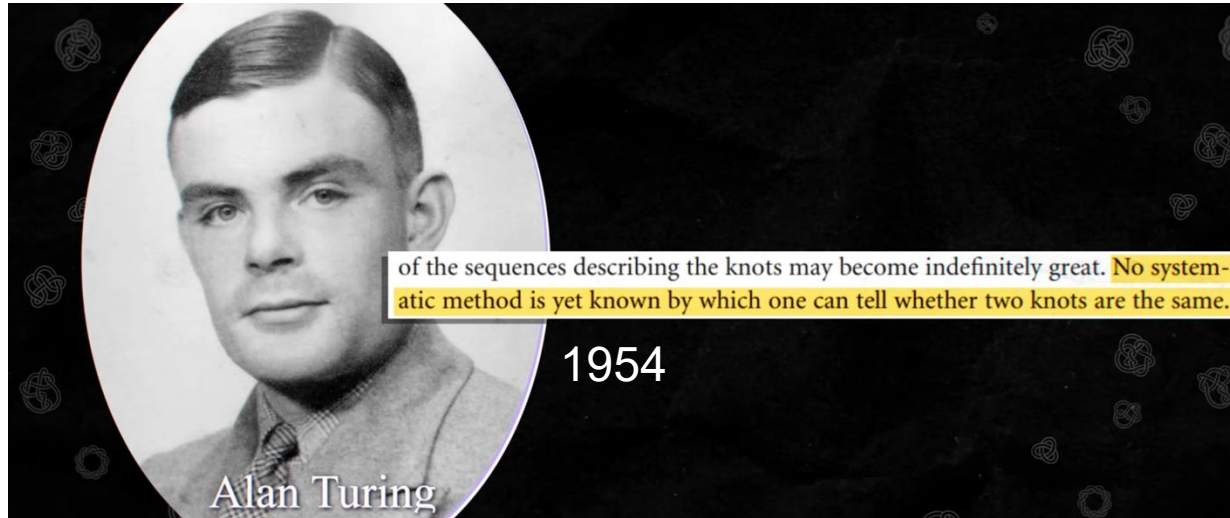


Michelson – Morley 1887 → no aether!

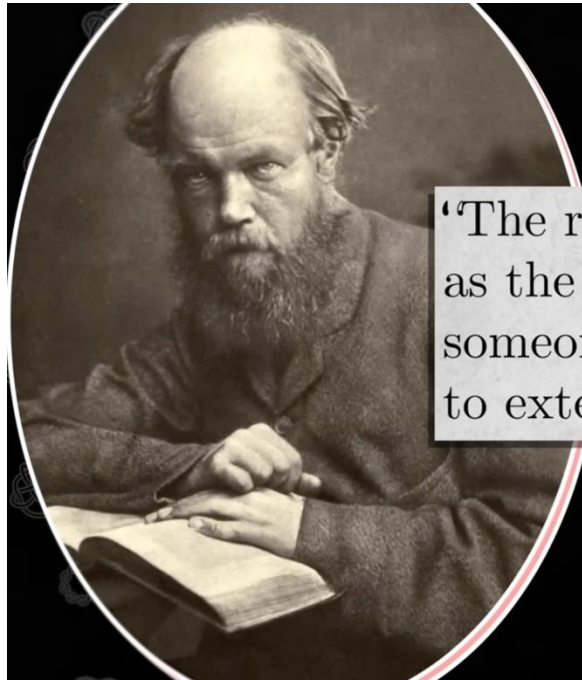
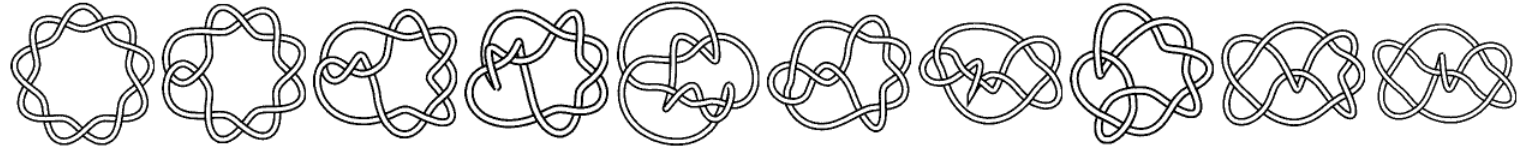


Knot OCD

Central Problem:
when are they
different?

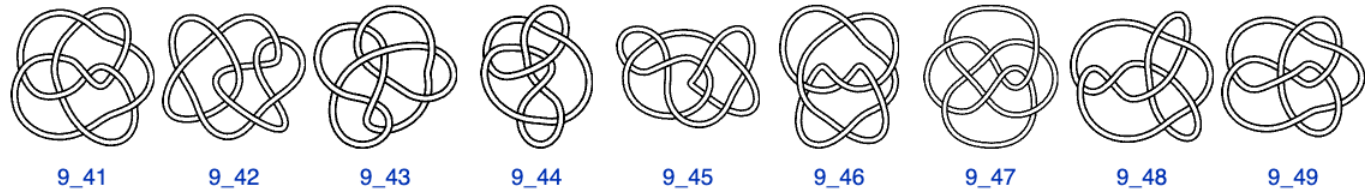
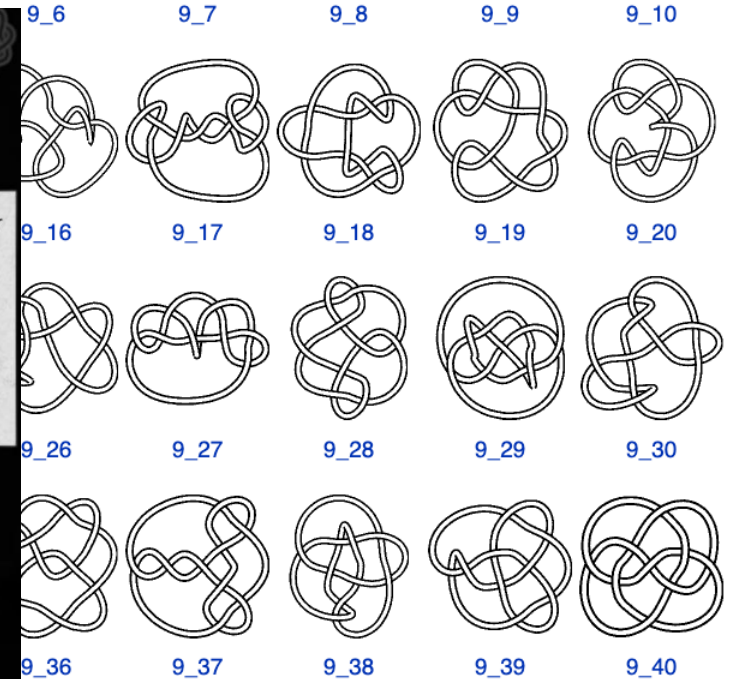


They did this by hand



“The requisite labour increases with extreme rapidity as the number of crossings is increased... someone with the requisite leisure should try to extend this list, if possible up to 11...”

Peter Tait (1831 – 1901)



Kirkman and Little continued Tait's tabulation

Topological Equivalence

Same Knot, Drawn Differently

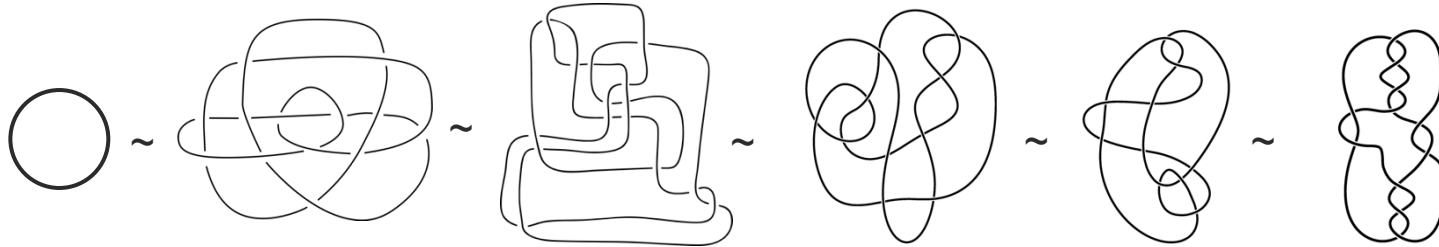


Reidemeister Move Sequence

~



These are all unknots! :



knot: one strand

link : more than one strand

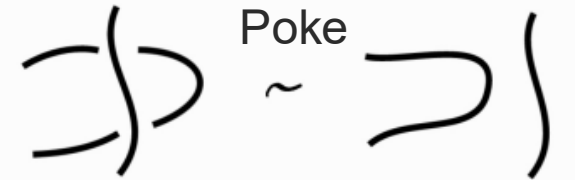


Reidemeister moves:

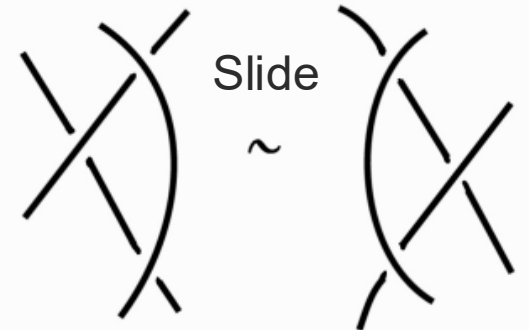
Twist



Poke

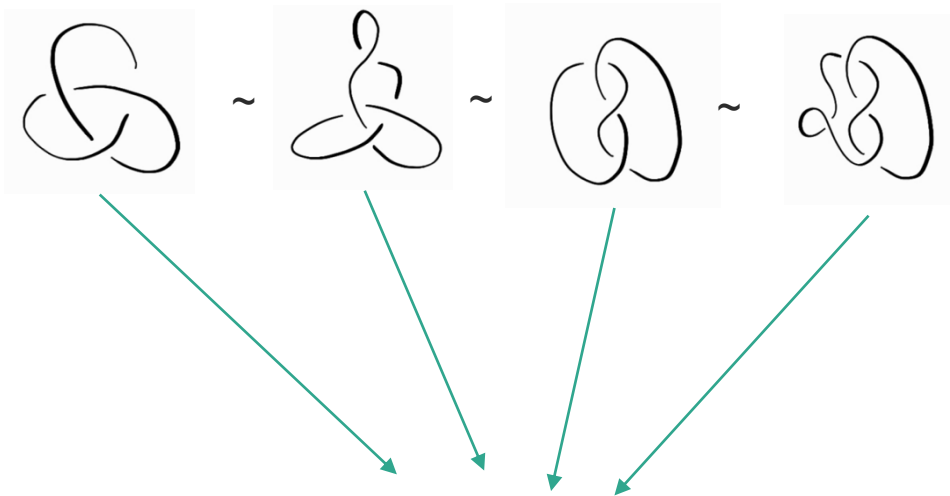


Slide



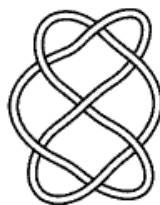
JONES POLYNOMIAL of an *oriented* link L : $V_L(t)$

Same Knot
(Reidemeister moves)



$$V_{3_1} = -t^{-4} + t^{-3} + t^{-1}$$

Same Polynomial



7_4

$$V_{7_4}(q) = -t^8 + t^7 - 2t^6 + 3t^5 - 2t^4 + 3t^3 - 2t^2 + t$$



7_5

Different Polynomial

$$V_{7_5}(t) = -t^{-9} + 2t^{-8} - 3t^{-7} + 3t^{-6} - 3t^{-5} + 3t^{-4} - t^{-3} + t^{-2}$$



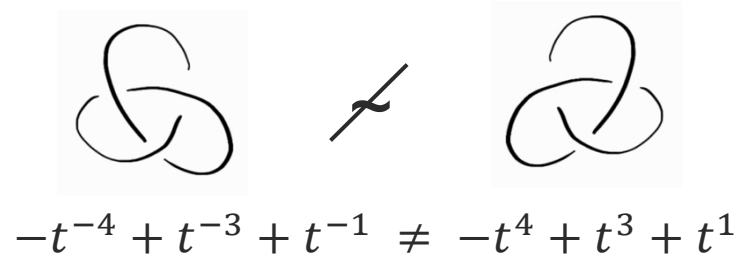
Different Knots

BUT! If two knots give same polynomial then we don't know.

Some properties

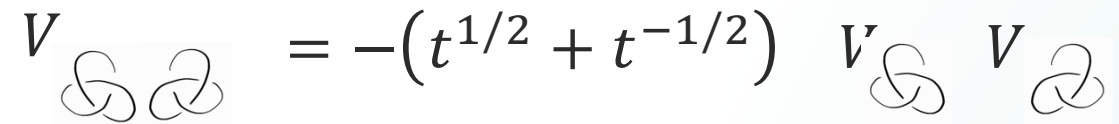
$$V_{\text{unknot}}(t) = 1$$

$$V_{\text{mirror}(L)}(t) = V_L(t^{-1})$$

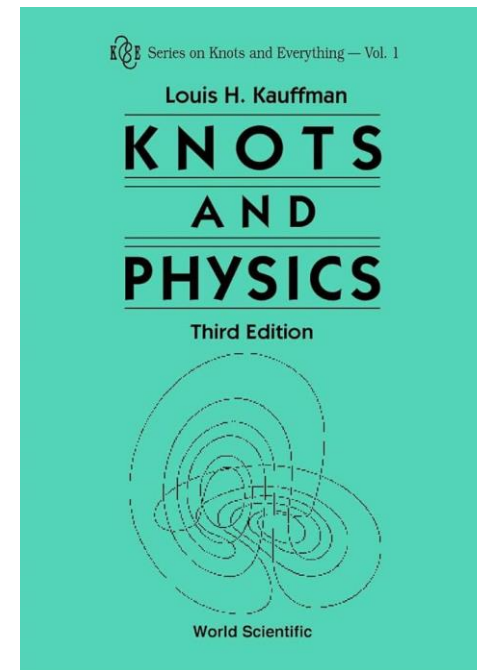


$$-t^{-4} + t^{-3} + t^{-1} \neq -t^4 + t^3 + t^1$$

$$V_{L_1 \sqcup L_2}(t) = -(t^{1/2} + t^{-1/2})V_{L_1}(t)V_{L_2}(t)$$



$$V_{\text{trefoil} \sqcup \text{link}} = -(t^{1/2} + t^{-1/2}) V_{\text{trefoil}} V_{\text{link}}$$



The Kauffman way

Kauffman Bracket

Crossing resolution

$$\langle \text{crossing} \rangle = A \langle \text{smooth} \rangle + A^{-1} \langle \text{smooth} \rangle$$

$$\langle \bigcirc D \rangle = (-A^2 - A^{-2}) \langle D \rangle \quad \text{Loop factor}$$

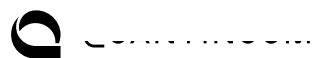
EXP BIG SUM 🦴 : $2^{\text{crossings}}$ terms

$$\begin{aligned} \langle D \rangle(A) &= \langle \text{trefoil} \rangle = A^3 \langle \text{smooth}_1 \rangle + A \langle \text{smooth}_2 \rangle + A \langle \text{smooth}_3 \rangle + A^{-1} \langle \text{smooth}_4 \rangle \\ &\quad + A \langle \text{smooth}_5 \rangle + A^{-1} \langle \text{smooth}_6 \rangle + A^{-1} \langle \text{smooth}_7 \rangle + A^{-3} \langle \text{smooth}_8 \rangle \\ &= (-A^2 - A^{-2})(-A^5 - A^{-3} + A^{-7}) \end{aligned}$$

(polynomial in A)

$$V_L(t) = \frac{(-A^3)^{-w(D)}}{(-A^2 - A^{-2})} \langle D \rangle(A), \text{ where } A = t^{-1/4} \quad \text{(now a polynomial in t)}$$

hard

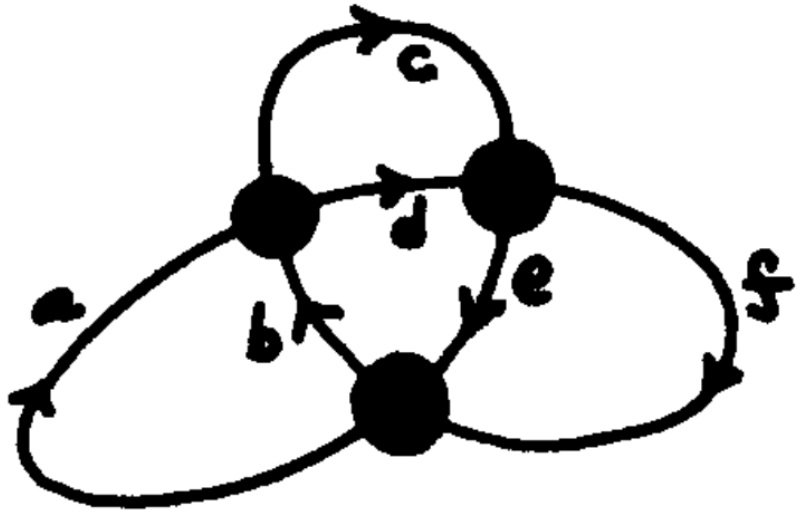


easy

Public

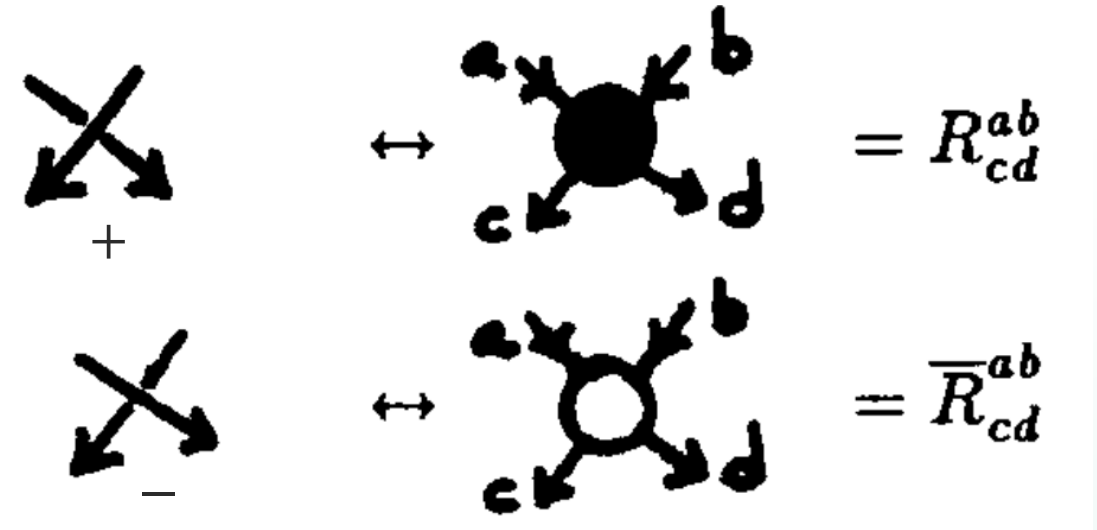
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Kauffman bracket as tensor network



$$= \sum_{a,b,c,d,e,f} R_{dc}^{ba} R_{ef}^{dc} R_{ba}^{ef}$$

Representing the exp big sum $\langle D \rangle$
as a tensor network
composed of two types of tensors



$$R = \begin{array}{c|ccc} A^{-1} & 0 & 0 & 0 \\ \hline 0 & -A^3 + A^{-1} & A & 0 \\ \hline 0 & A & 0 & 0 \\ \hline 0 & 0 & 0 & A^{-1} \end{array}$$

$$\bar{R} : A \leftrightarrow A^{-1}$$

ABSTRACT TENSOR CONTRACTION

<https://github.com/tlaakkonen/jones-tensor>

Representing the exp big sum $\langle D \rangle$
as a tensor network.

We can now use **tensor contraction** heuristics
to achieve **sota** for 1000s of crossings.

Hyper-optimized tensor network contraction

Johnnie Gray^{1,2} and Stefanos Kourtis^{1,3,4}

¹Blackett Laboratory, Imperial College London, London SW7 2AZ, United Kingdom

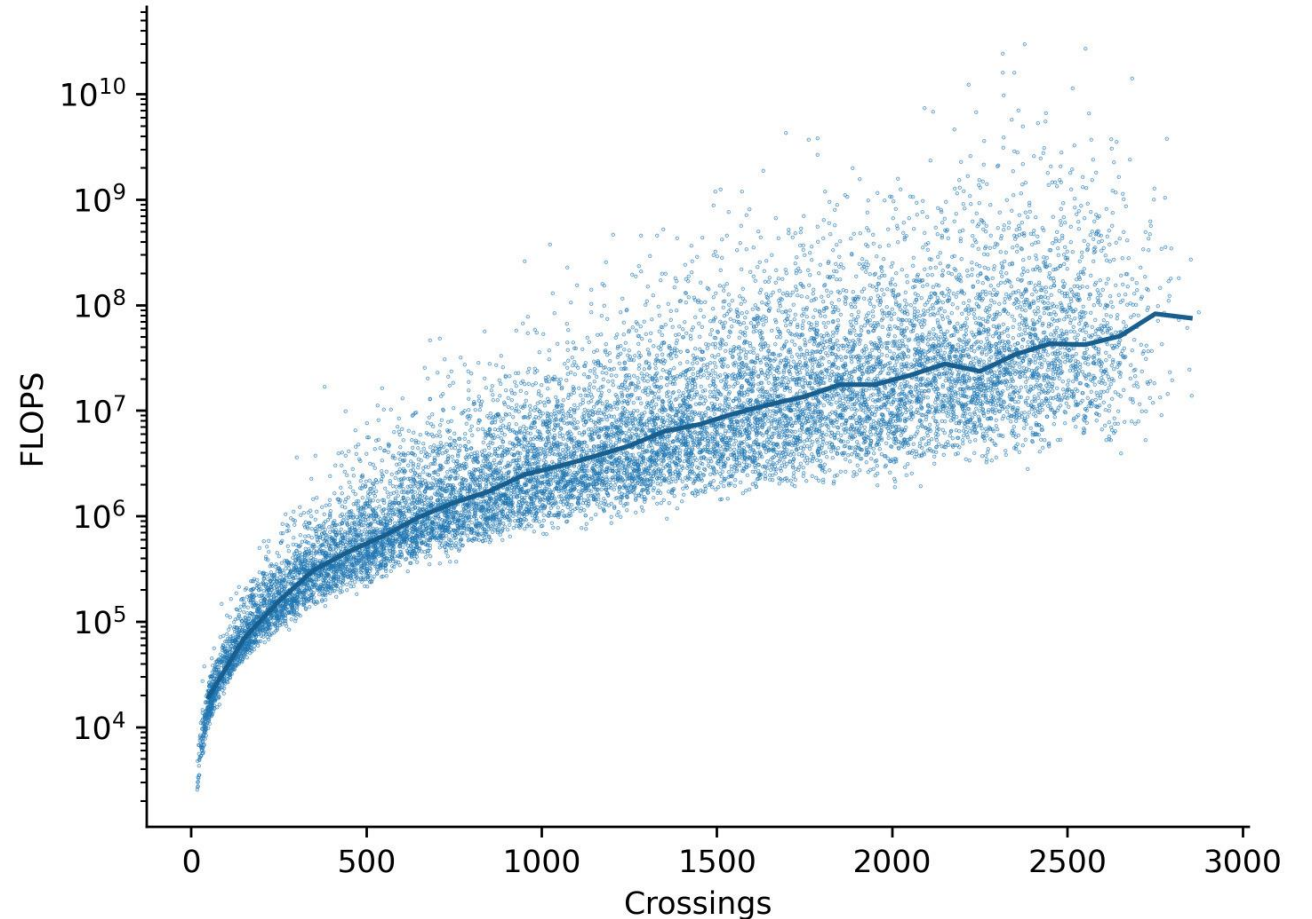
²Division of Chemistry and Chemical Engineering, California Institute of Technology, Pasadena, California 91125, USA

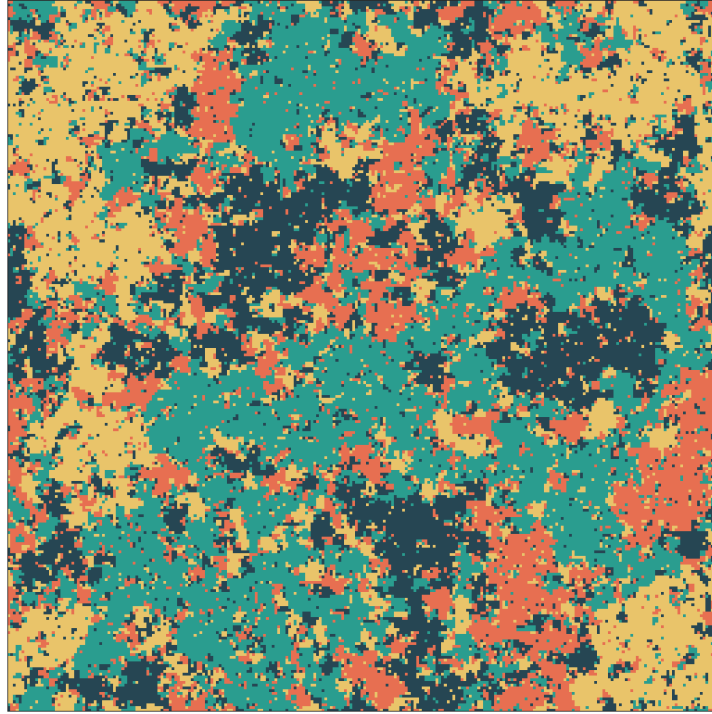
³Department of Physics, Boston University, Boston, MA, 02215, USA

⁴Institut quantique & Département de physique, Université de Sherbrooke, Québec J1K 2R1, Canada

March 12, 2021

Contraction Path-finding for Random Links



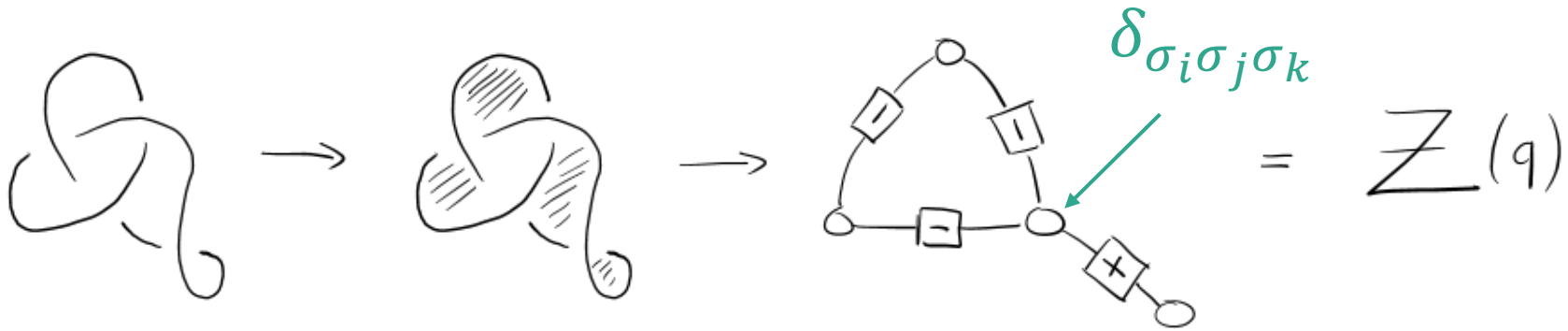


The Potts way

Jones from Potts model partition function

$$\mathcal{Z}_K(q) = \sum_{\{\sigma\}} \prod_{(i,j)} e^{\delta_{\sigma_i \sigma_j} J_{ij}} \in \mathbb{C}$$

$$e^{J_{\pm}} = -t^{\mp 1} \quad q = (t^{1/2} + t^{-1/2})^2$$



$$\epsilon_i - \boxed{\pm} - \epsilon_j = 1 - (1 + t^{\mp 1}(q)) \delta_{\epsilon_i \epsilon_j} = \begin{cases} -t^{\mp 1}(q), & \epsilon_i = \epsilon_j \\ 1, & \text{otherwise} \end{cases}$$

$$t(q) = \frac{1}{2}(q + \sqrt{q} \sqrt{q-4} - 2)$$

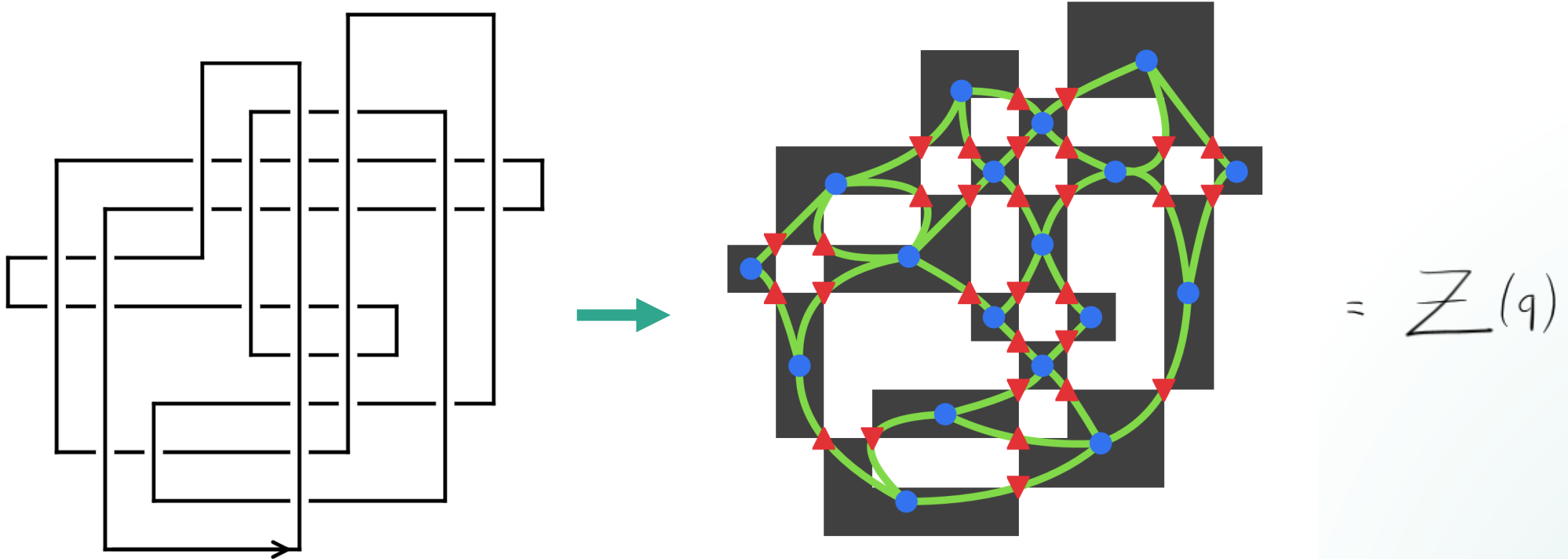
$$V_K(t(q)) = (-t^{\frac{1}{2}} - t^{-\frac{1}{2}})^{-(n+1)} (-t^{3/4})^w t^{\tau/4} \mathcal{Z}_K(q)$$

easy

hard

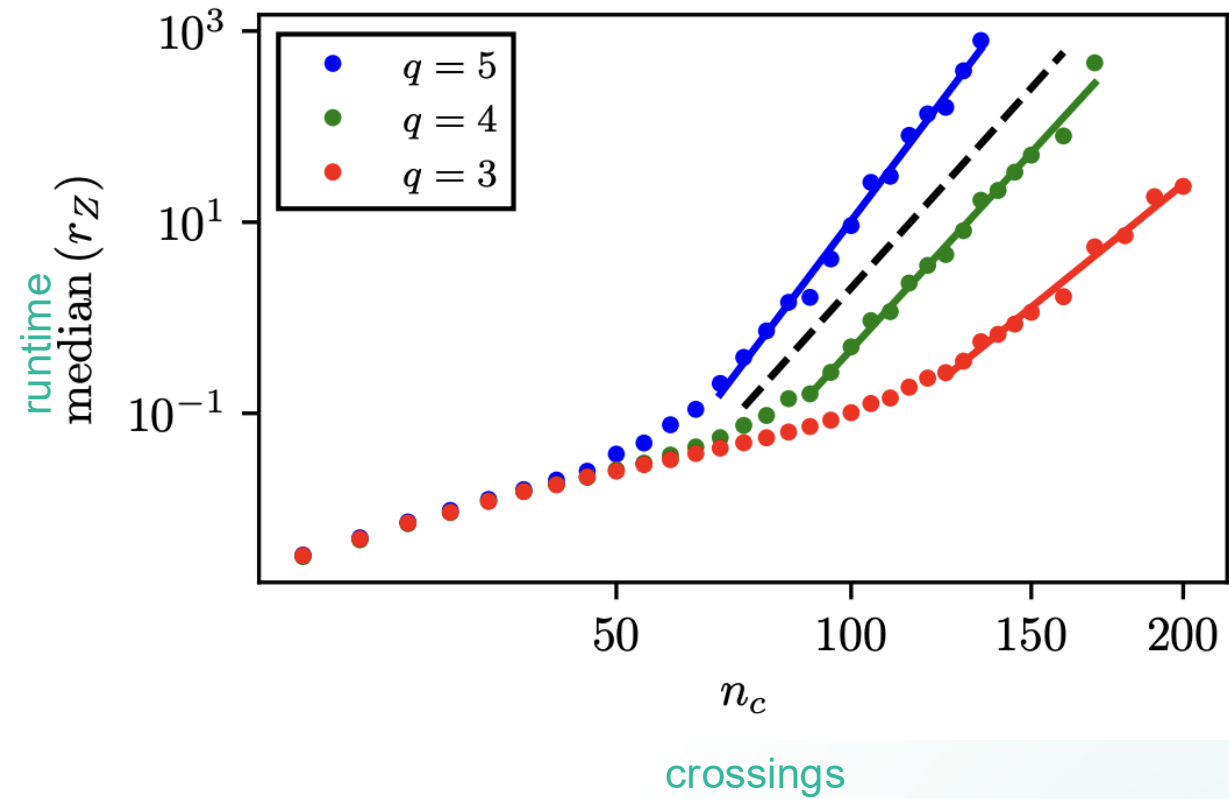
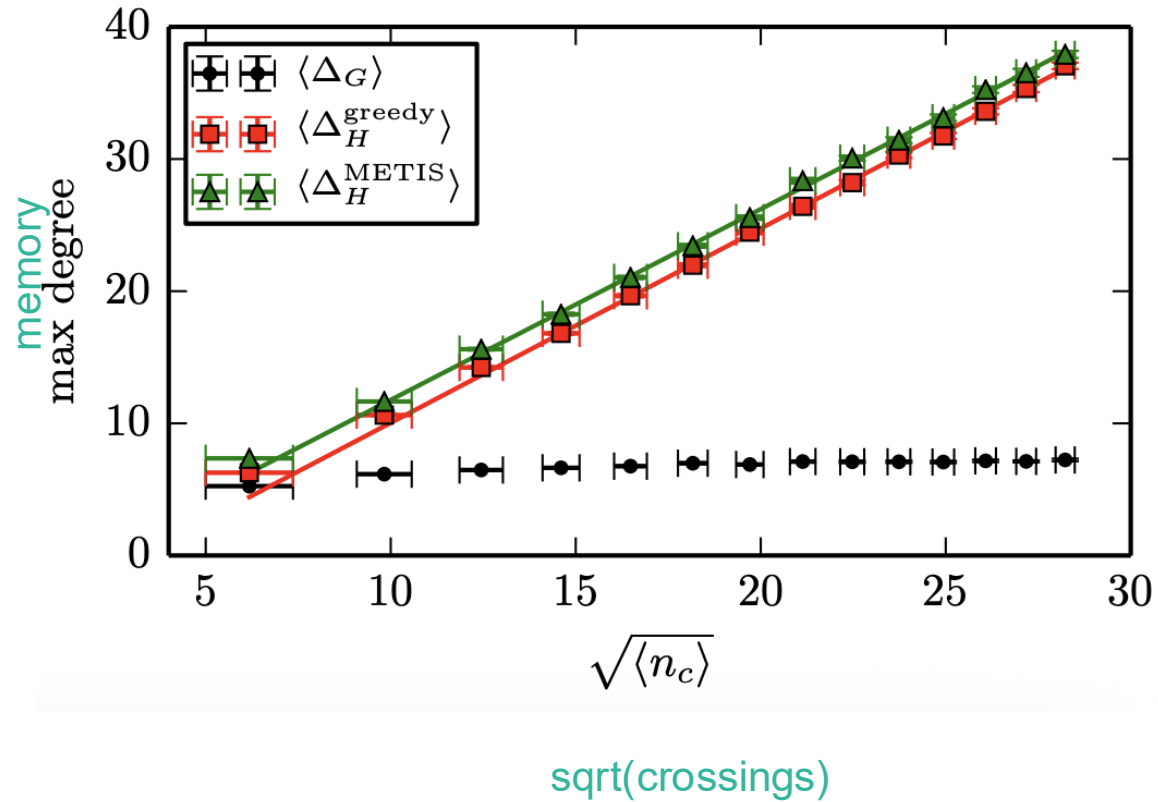
Potts partition function as tensor network

#P-hard in general



Subexponential contraction cost

Planarity : sqrt on exponent





I guess, bro ...

where is the k w a n t o m ?

QUANTUM

[nature](#) > [news](#) > article

NEWS | 10 April 2025 | Correction [11 April 2025](#)

‘Mind blowing’: quantum computer untangles the mathematics of knots

Algorithms for studying knots and other topological objects could have a quantum advantage.

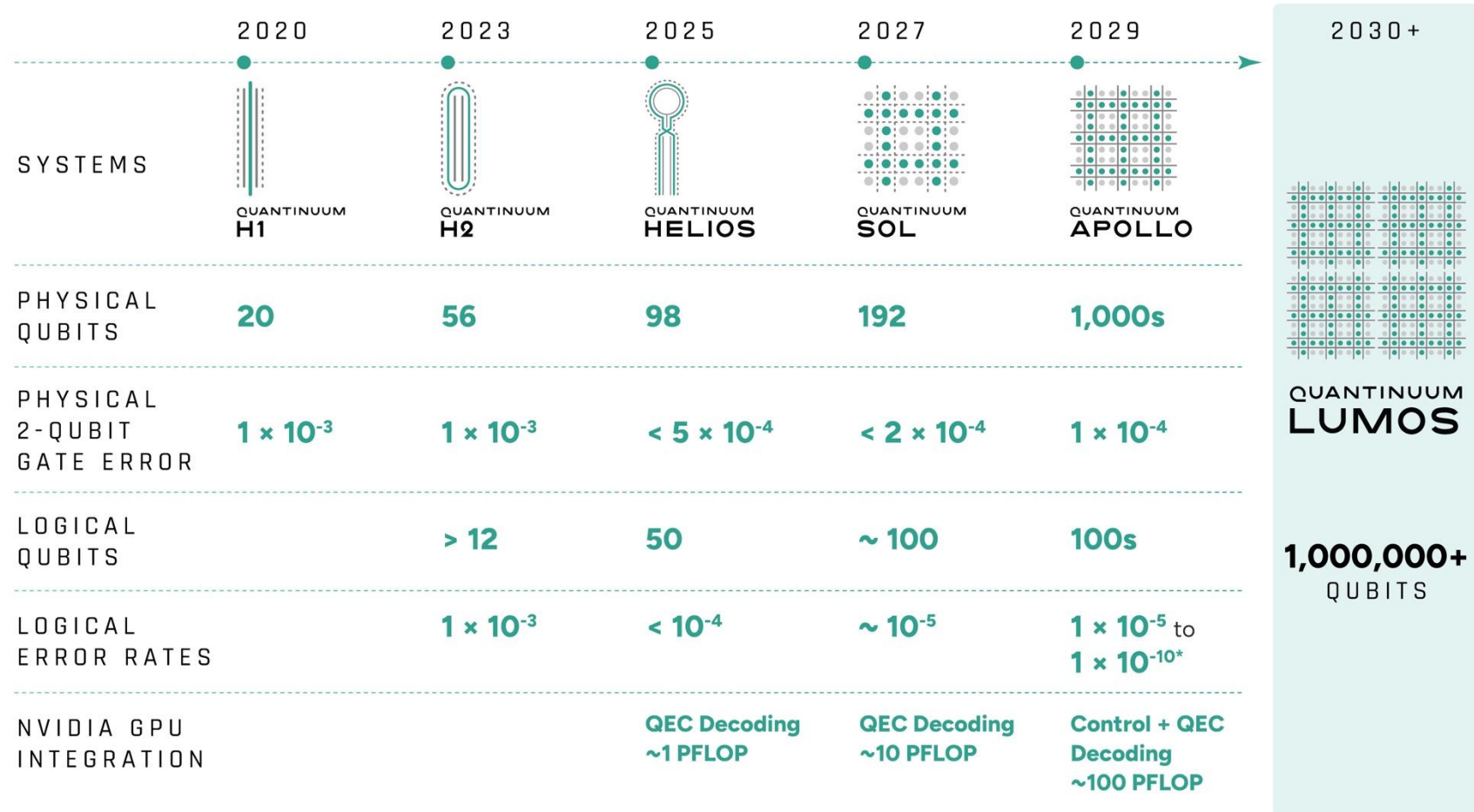
By [Davide Castelvecchi](#)



You are here



QNTM HW Roadmap



*analysis based on recent literature in new, novel error correcting codes predict that error could be as low as $1E-10$ in Apollo (ref: arXiv:2403.16054, arXiv:2308.07915).

HELIOS

Specs

98 fully-connected physical qubits

99.921%
two-qubit gate fidelity

99.9975%
single-qubit gate fidelity

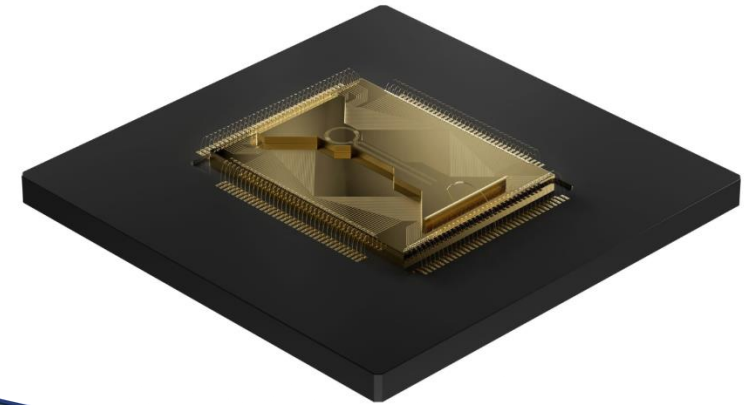
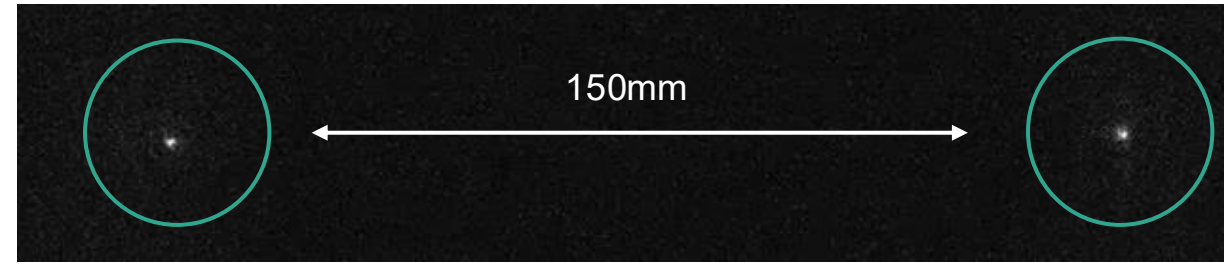
Surface electrode ion trap architecture (QCCD)

Mid-Circuit Measurement
Conditional Classical Logic

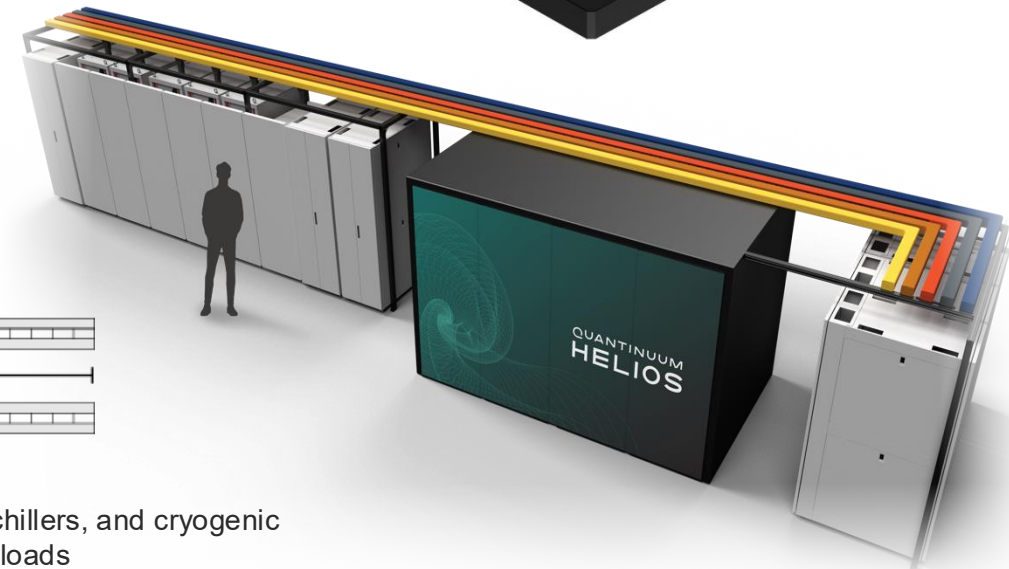
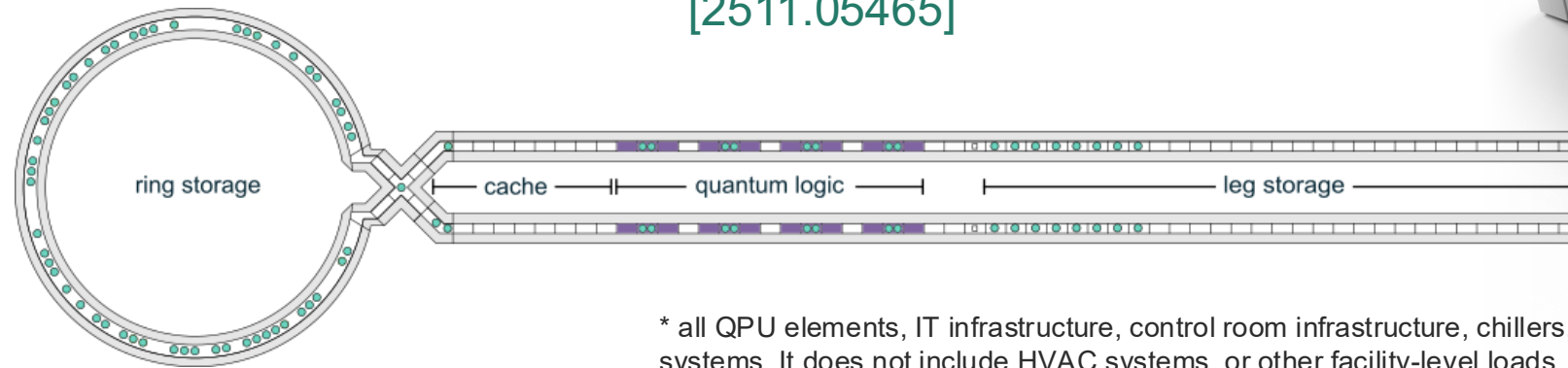
50 ms depth-1 time

~60kW power consumption*

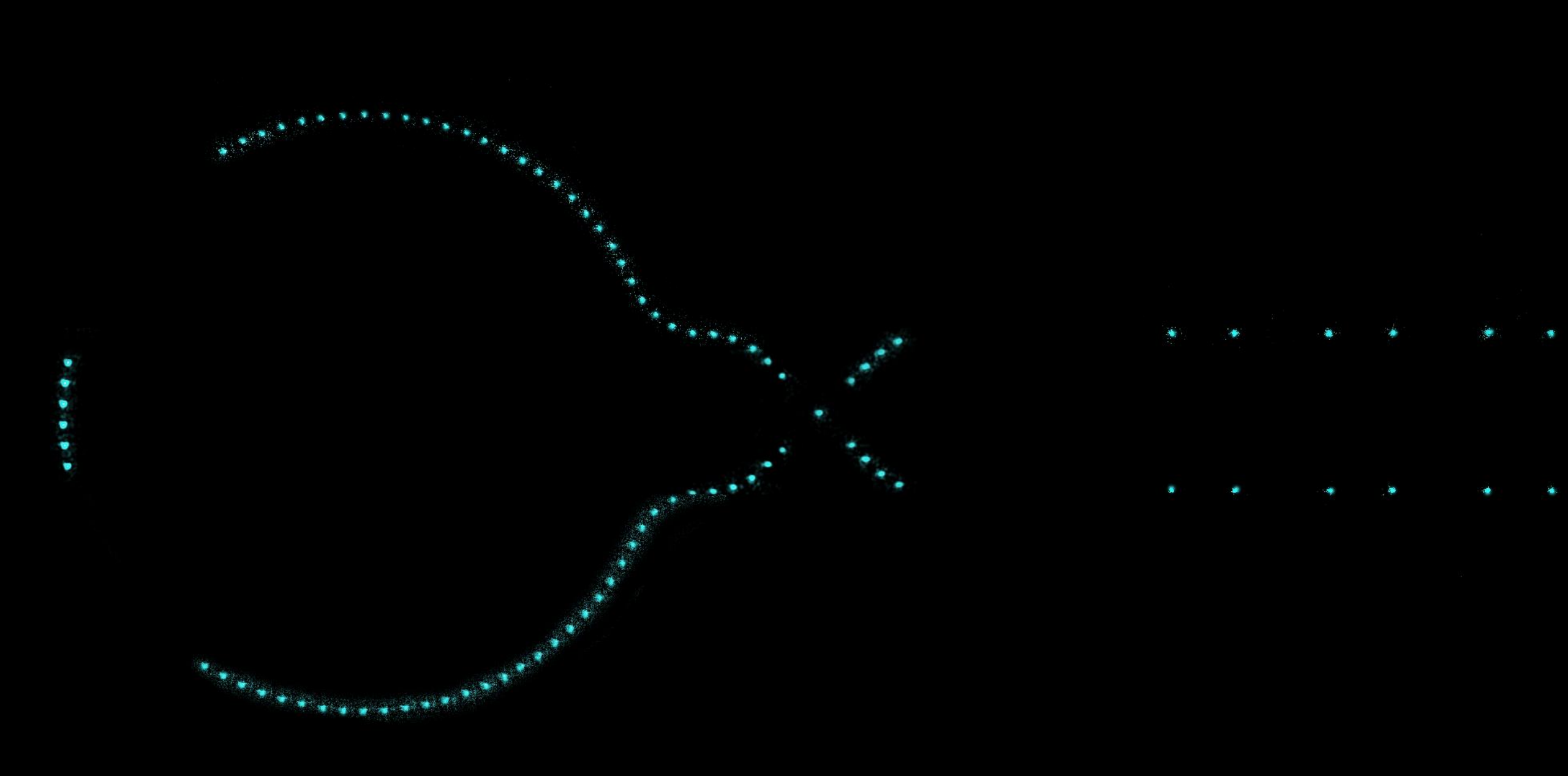
Split and Combine



[2511.05465]

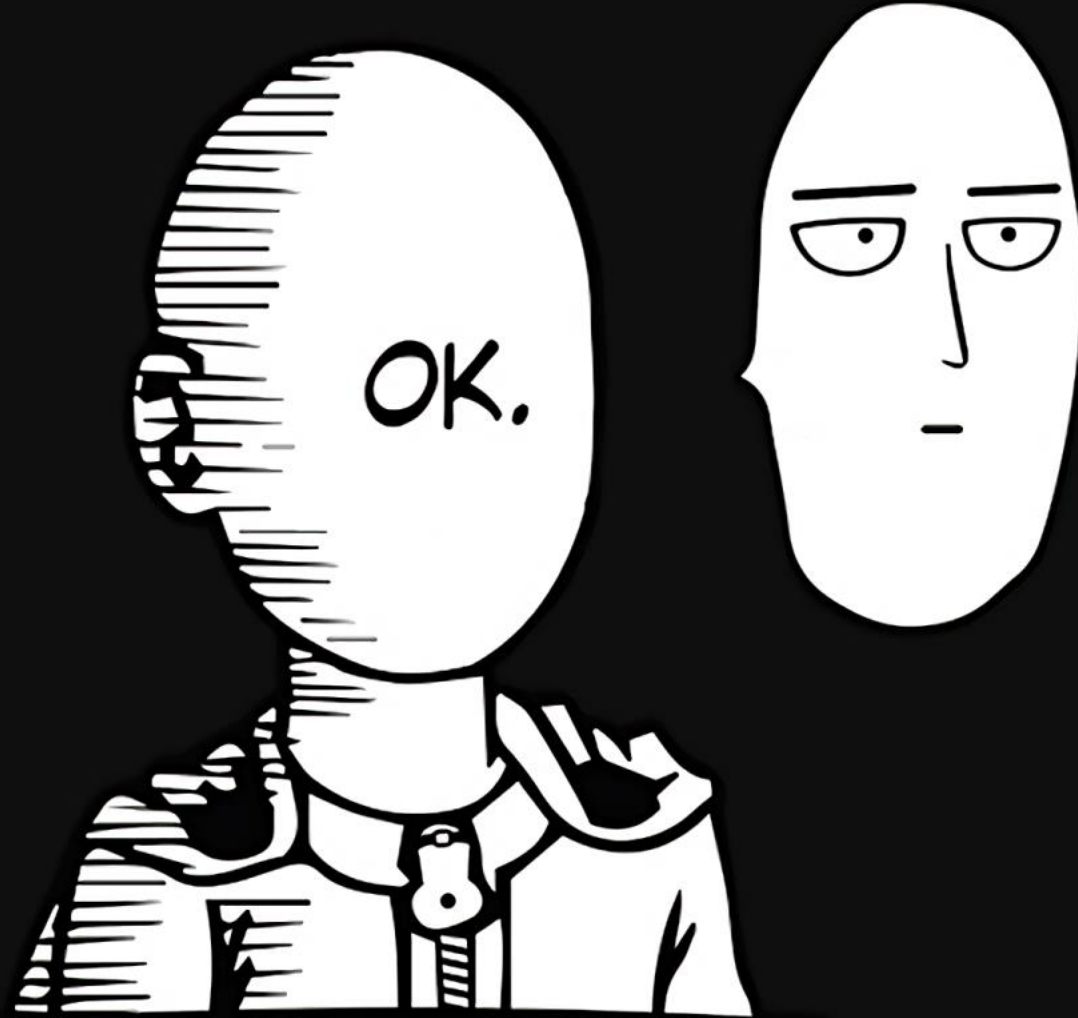


* all QPU elements, IT infrastructure, control room infrastructure, chillers, and cryogenic systems. It does not include HVAC systems, or other facility-level loads



Atomic ions (single atoms) imaged inside the Helios QPU

How do we put a link in a quantum computer ???



Alexander thm: Every link is a closed Braid.

Braid generators:

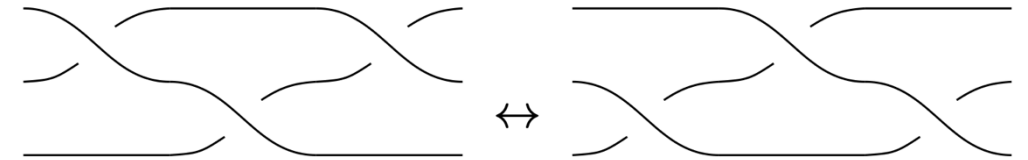
$$\sigma_i = \begin{matrix} i \\ i+1 \end{matrix} \begin{array}{c} \text{---} \backslash \\ \text{---} / \end{array}$$

$$\sigma_i^{-1} = \begin{matrix} i \\ i+1 \end{matrix} \begin{array}{c} \text{---} / \\ \text{---} \backslash \end{array}$$

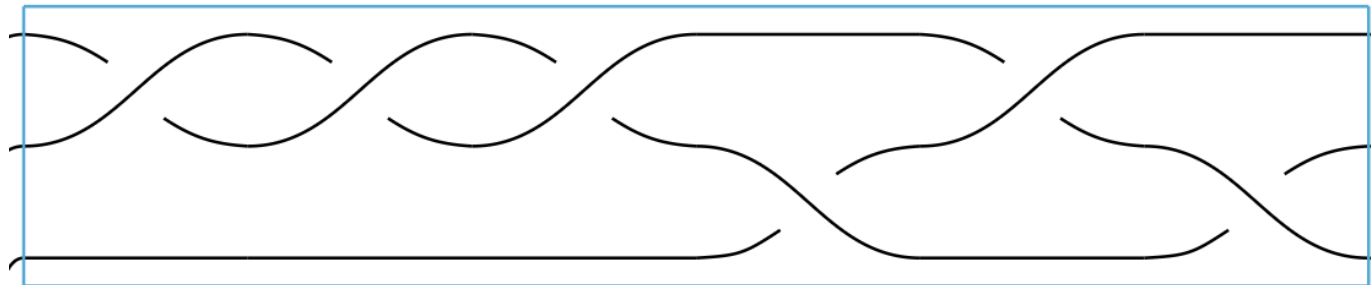
$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad \text{for } |i - j| \geq 2$$



$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \quad \text{for } 1 \leq i \leq n - 2$$



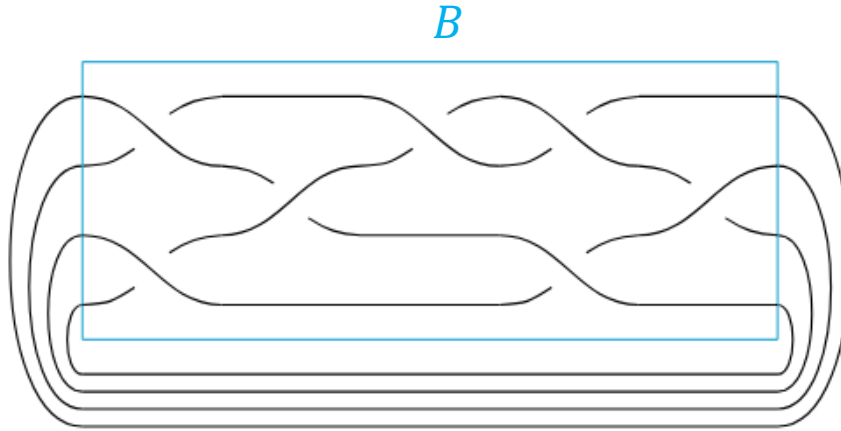
$$B = (\sigma_1^{-1})^3 \sigma_2 \sigma_1^{-1} \sigma_2$$



Two ways to close a braid B :

Markov

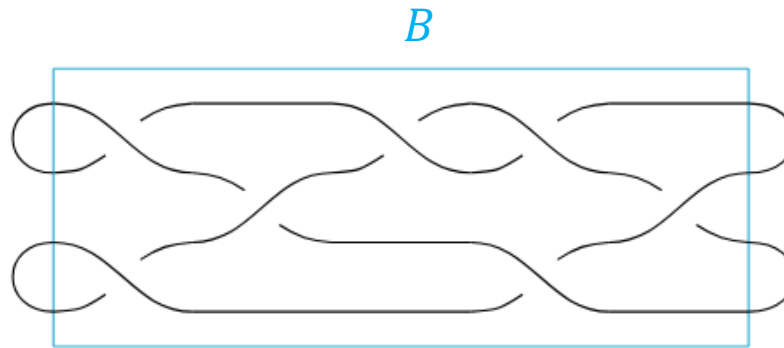
$M(B) =$



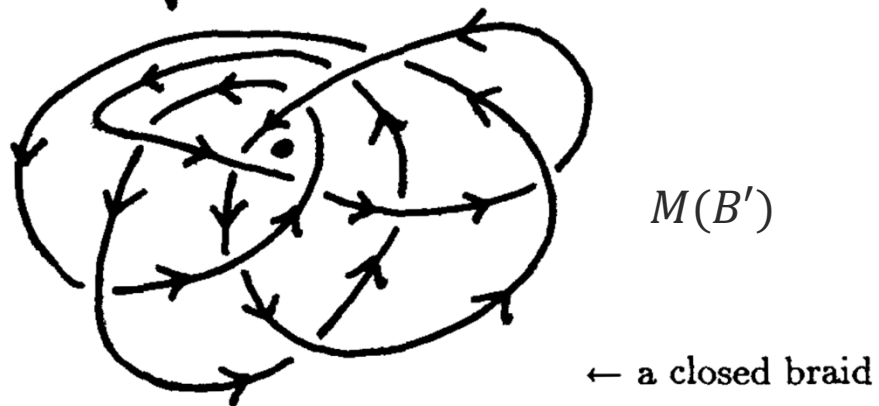
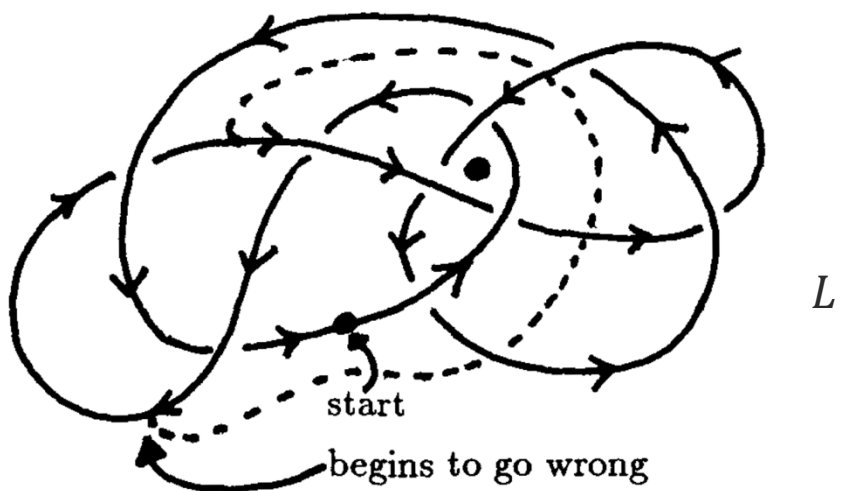
“Trace”

Plat

$P(B) =$

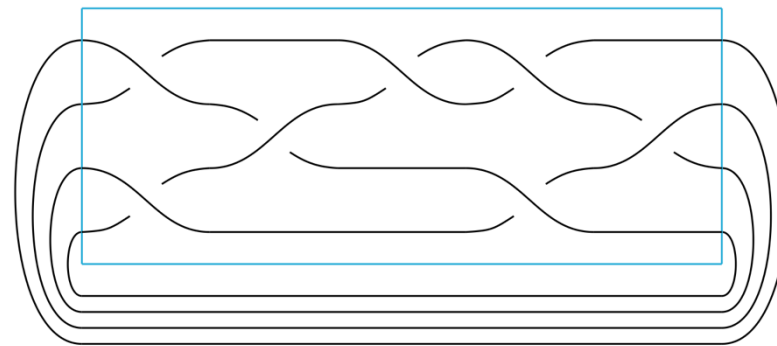


“Cup and Caps”

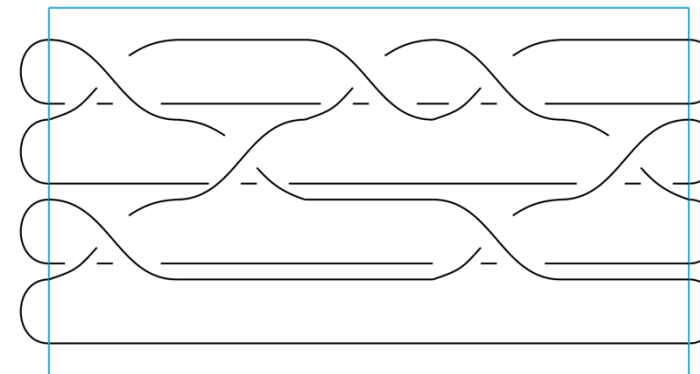


Vogel's algorithm: given a link $P(B)$, it's easy to get a braid B' such that $M(B') = P(B)$.
 B' won't be optimal (in terms of strands or crossings).

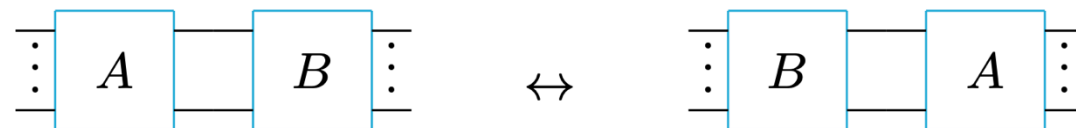
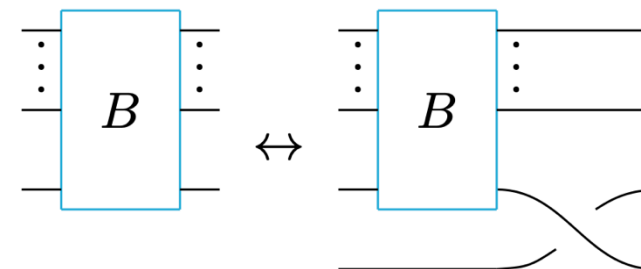
$$M(B) =$$



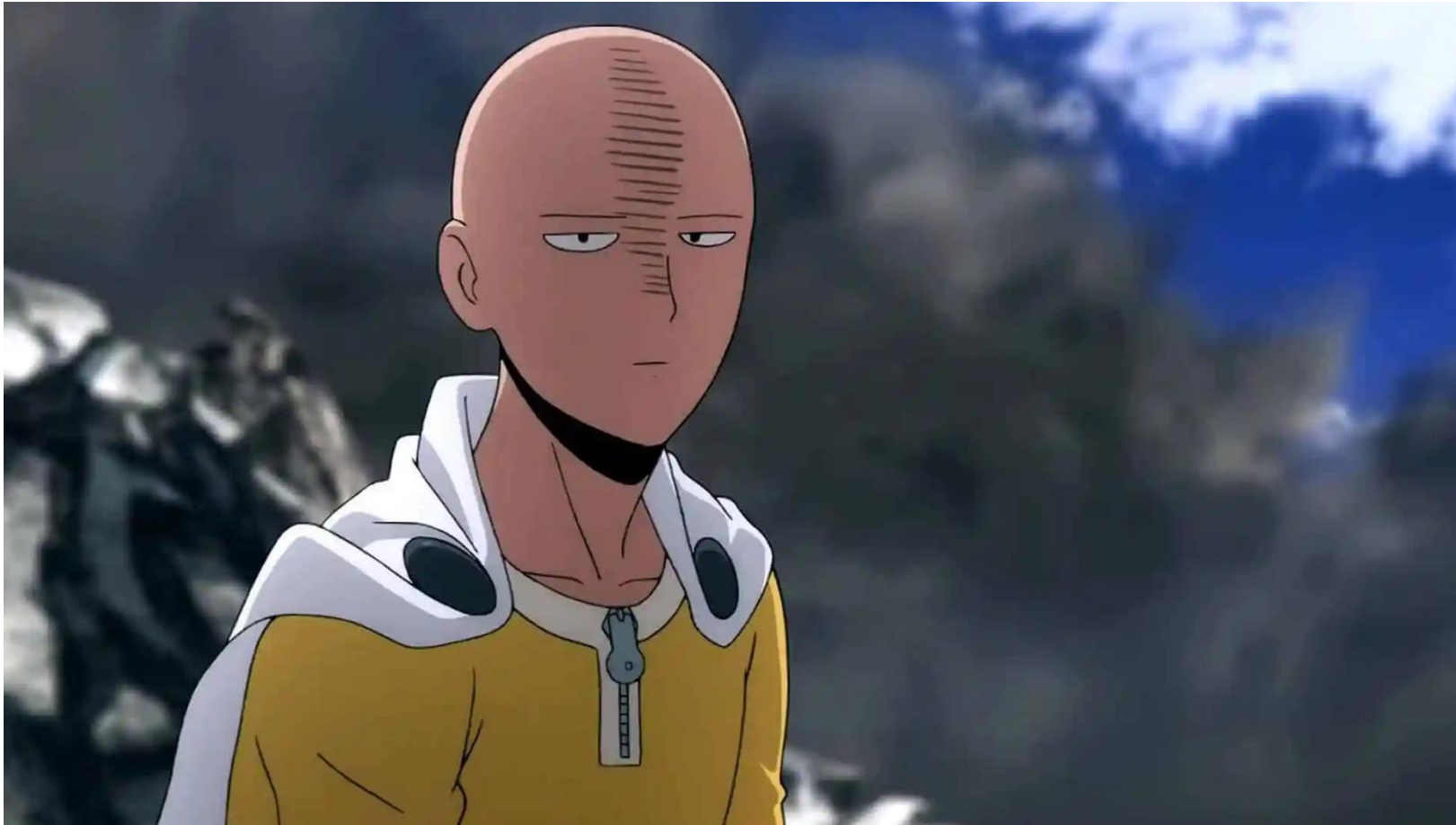
$$\text{slide} \sim P(B') =$$



Markov moves (leave Markov-closed braid invariant):



So.... How do we put a ***braid*** in a quantum computer... ???



Ah! and what does closure look like as a quantum operation ?

BRAIDS HAVE UNITARY REPRESENTATIONS!

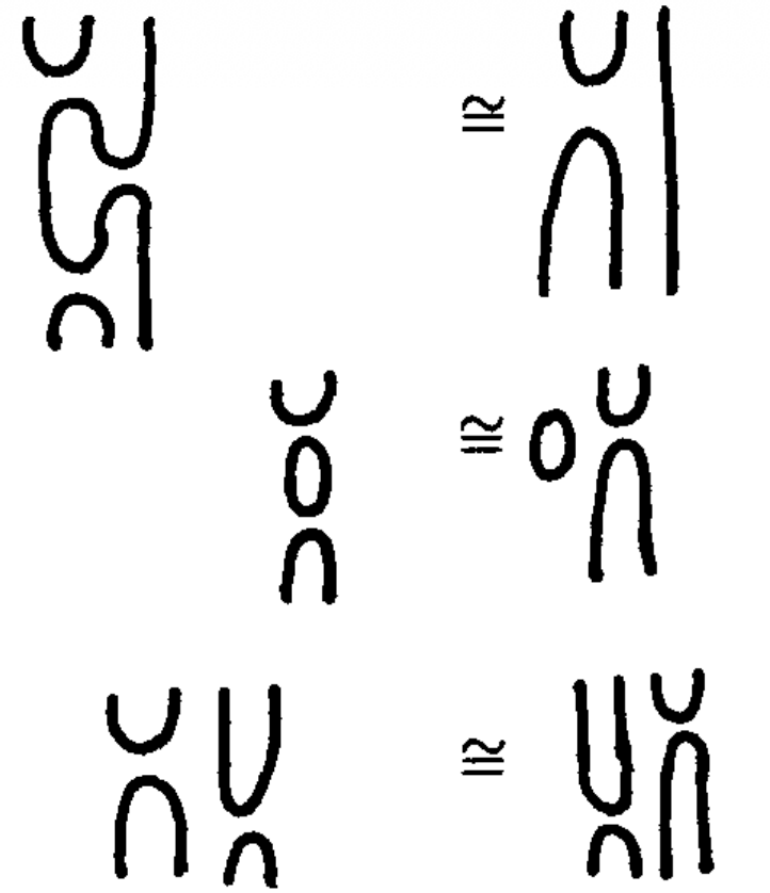
$$\langle \text{X} \rangle = A \langle \text{)(} \rangle + A^{-1} \langle \text{U} \rangle$$

$\sigma_i = A \cdot I + A^{-1} \cdot U_i \xrightarrow{\text{Unitarity}} A = e^{i\theta}$

$$U_i U_{i\pm 1} U_i = U_i$$

$$U_i^2 = (-A^2 - A^{-2}) U_i$$

$$U_i U_j = U_j U_i \text{ for } |i - j| > 1$$



Evaluating the Jones at roots of unity is **complete** for **two** quantum complexity classes

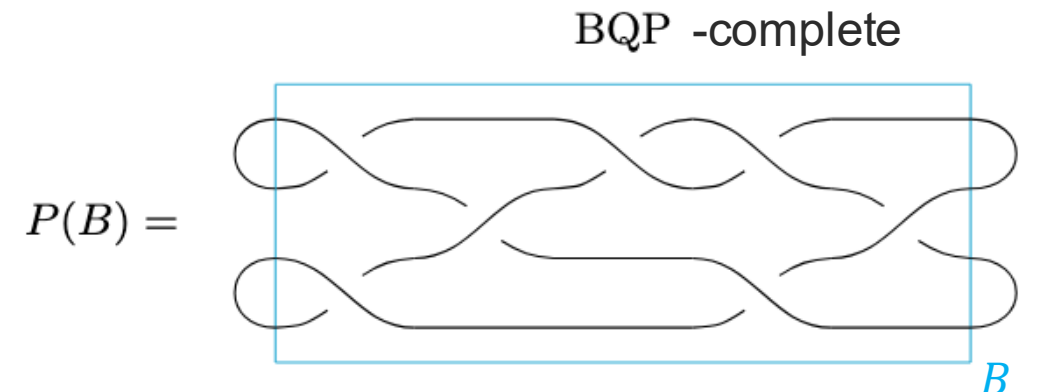
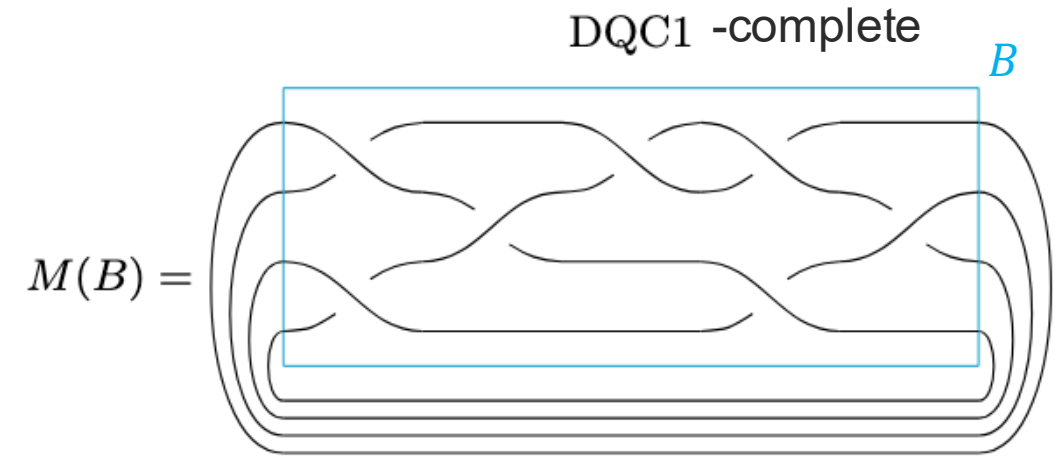
At $t = e^{i\theta}$, $\theta \in [0, 2\pi]$ the problem is quantum-easy and classically-hard in the worst case.

Exception: 'lattice roots of unity' where it is classically easy: $\{\exp(i\theta) \mid \theta = \pm\pi, \pm\pi/2, \pm\pi/3, \pm 2\pi/3\}$

Let's focus on $e^{i\frac{2\pi}{5}}$, the “Fibonacci” point.

Problem. *APPROX-JONES-MARKOV: Given a braid B on $n - 1$ strands and $c = O(\text{poly}(n))$ crossings, compute $R \in \mathbb{C}$ such that $|V_{M(B)}(e^{i\frac{2\pi}{5}}) - R| \leq \epsilon \phi^{n-1}$ with high probability, where $\epsilon^{-1} = O(\text{poly}(n))$.*

Problem. *APPROX-JONES-PLAT: Given a braid B on $n - 1$ strands and $c = O(\text{poly}(n))$ crossings, compute $R \in \mathbb{C}$ such that $|V_{P(B)}(e^{i\frac{2\pi}{5}}) - R| \leq \epsilon \phi^{\frac{n}{2}}$ with high probability, where $\epsilon^{-1} = O(\text{poly}(n))$.*



* This does not imply $\text{DQC}_1 = \text{BQP}$



QUANTINUUM

Converting from Plat to Markov introduces exp samples (Shor)

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Representation at the 5th root of unity

$A = e^{3\pi i/5} \rightarrow t = e^{i\frac{2\pi}{5}}$

$U_{\sigma_i} =$

	$ 000\rangle$	$ 001\rangle$	$ 100\rangle$	$ 010\rangle$	$ 011\rangle$	$ 110\rangle$	$ 101\rangle$	$ 111\rangle$
$ 000\rangle$	1	0	0	0	0	0	0	0
$ 001\rangle$	0	1	0	0	0	0	0	0
$ 100\rangle$	0	0	1	0	0	0	0	0
$ 010\rangle$	0	0	0	$-A^{-3}$	0	0	0	0
$ 011\rangle$	0	0	0	0	A	0	0	0
$ 110\rangle$	0	0	0	0	0	A	0	0
$ 101\rangle$	0	0	0	0	0	0	$\phi^{-1}e^{i4\pi/5}$	$\phi^{-1/2}e^{i7\pi/5}$
$ 111\rangle$	0	0	0	0	0	0	$\phi^{-1/2}e^{i7\pi/5}$	$-\phi^{-1}$

Where i indexes the middle qubit

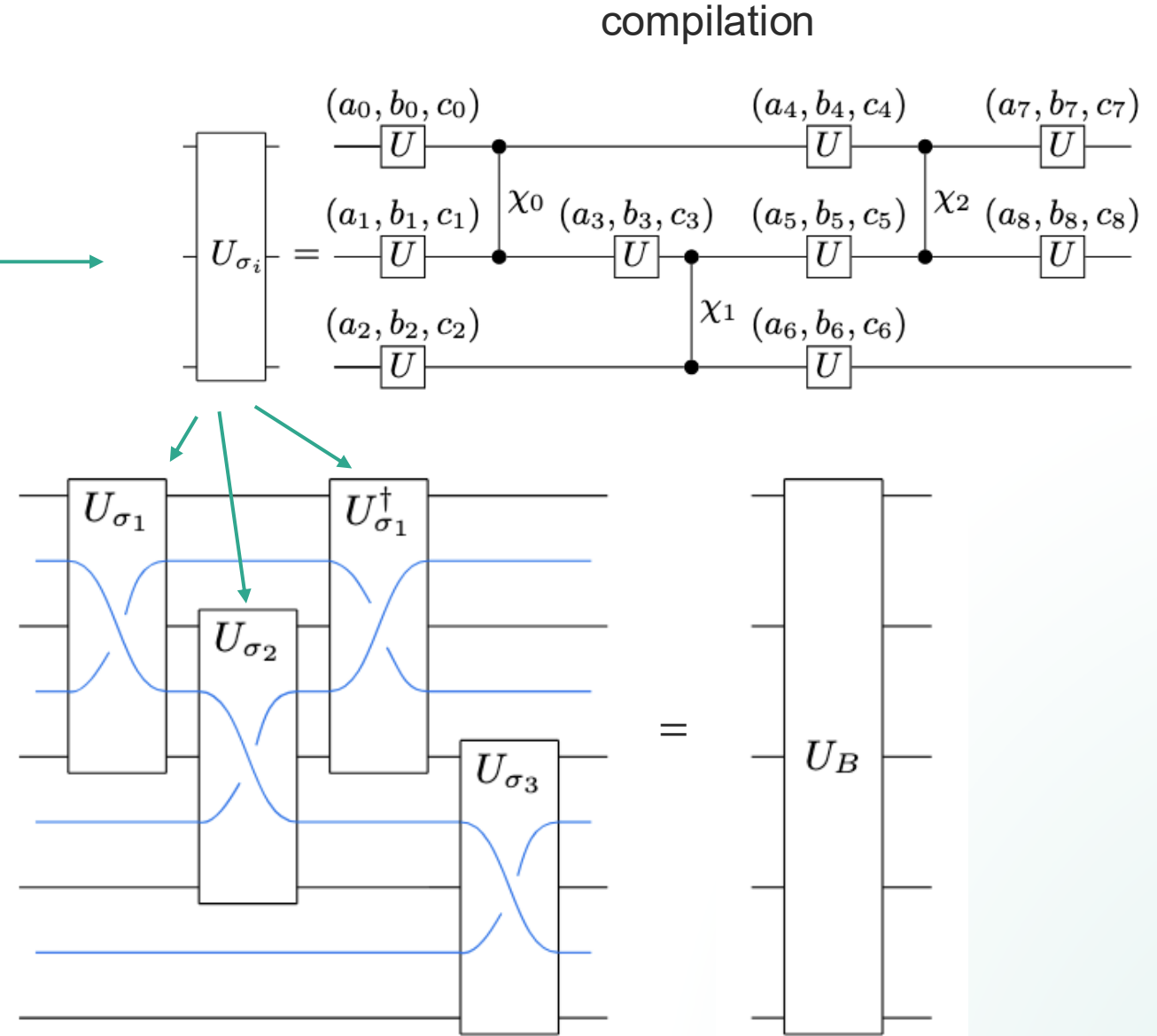
“Fibonacci point” : basis constrained to have **no consecutive zeros**
Number of bistrings satisfying this constraint is always a Fibonacci number

Compilation at $t = e^{i\frac{2\pi}{5}}$

Unitary representation of σ_i (Kauffman):

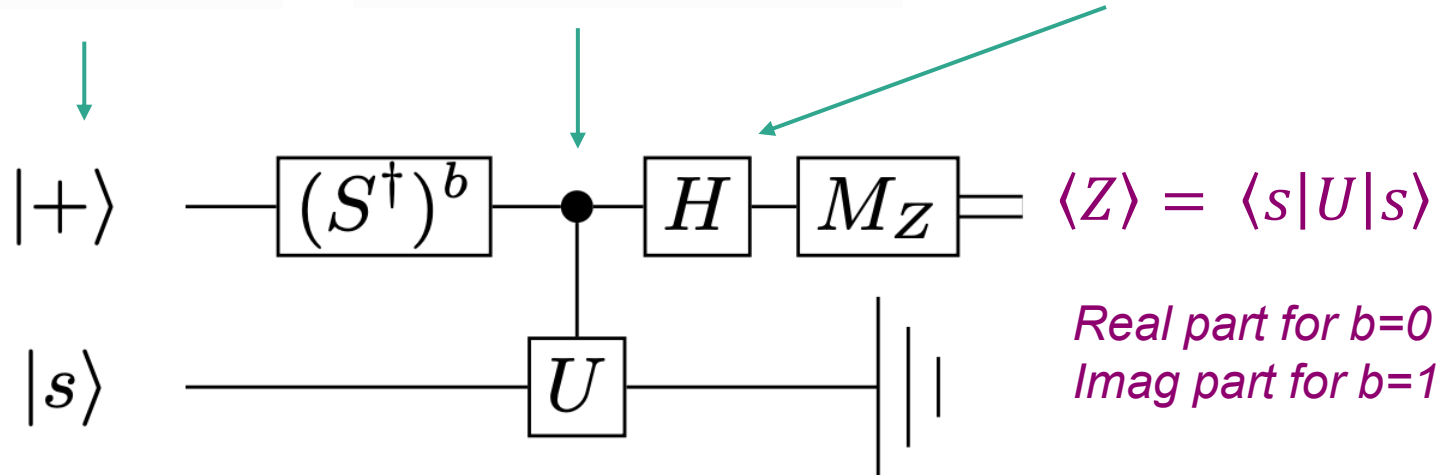
$$\begin{aligned}\langle 101|U_{\sigma_i}|101\rangle &= \phi^{-1}e^{i\frac{4\pi}{5}} \\ \langle 111|U_{\sigma_i}|111\rangle &= -\phi^{-1} \\ \langle 010|U_{\sigma_i}|010\rangle &= e^{-i\frac{4\pi}{5}} \\ \langle 011|U_{\sigma_i}|011\rangle &= e^{i\frac{3\pi}{5}} \\ \langle 110|U_{\sigma_i}|110\rangle &= e^{i\frac{3\pi}{5}} \\ \langle 111|U_{\sigma_i}|101\rangle &= \phi^{-\frac{1}{2}}e^{-i\frac{3\pi}{5}} \\ \langle 101|U_{\sigma_i}|111\rangle &= \phi^{-\frac{1}{2}}e^{-i\frac{3\pi}{5}}\end{aligned}$$

A braid is represented as a **unitary matrix** which is then compiled as a **quantum circuit** (every crossing is a 3-qubit gate)



Key algorithmic primitive: Hadamard test

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |s\rangle \quad \frac{1}{\sqrt{2}}(|0\rangle \otimes |s\rangle + |1\rangle \otimes U|s\rangle) \quad \frac{1}{2}|0\rangle \otimes (|s\rangle + U|s\rangle) + \frac{1}{2}|1\rangle \otimes (|s\rangle - U|s\rangle)$$



$$P(0) = \frac{1}{2}(1 + \text{Re}\langle s|U|s\rangle)$$

$$P(1) = 1 - P(0) = \frac{1}{2}(1 - \text{Re}\langle s|U|s\rangle)$$

$$\langle Z \rangle = P(0) - P(1) = \text{Re}\langle s|U|s\rangle$$

Similarly for imaginary part

And if I uniformly randomly sample the bitstring $|s\rangle$ (maximally mixed state):

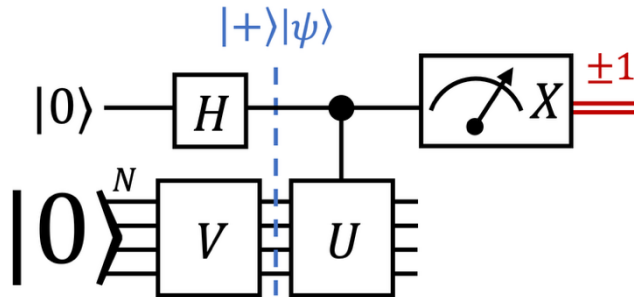
$$\text{tr}(U) = \sum_{s=0}^{2^n-1} \langle s|U|s\rangle$$

$$\mathbb{E}[Z] = \frac{1}{2^n} \text{Re} \left(\sum_{s=0}^{2^n-1} \langle s|U|s\rangle \right) = \frac{1}{2^n} \text{Re}(\text{tr}(U))$$

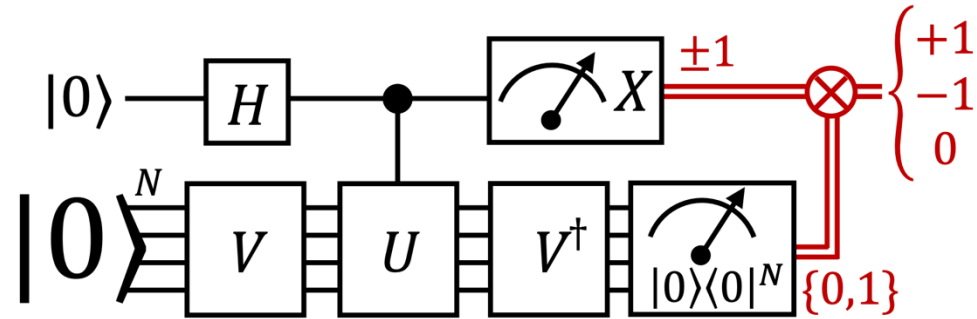
→ we'll use a fancier version of H-test

Tricks: removing the ancilla and using echo verification [2207.09479]

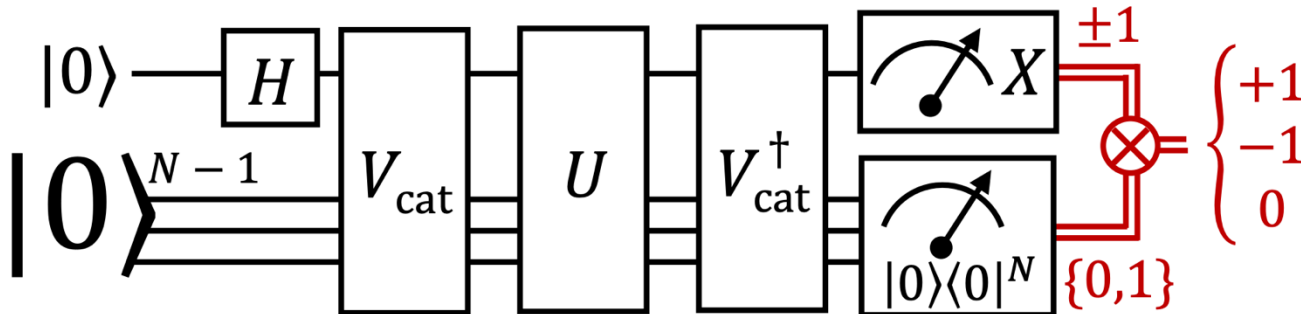
Vanilla H-test:



Echo-verification:



Control-free Echo Verification:



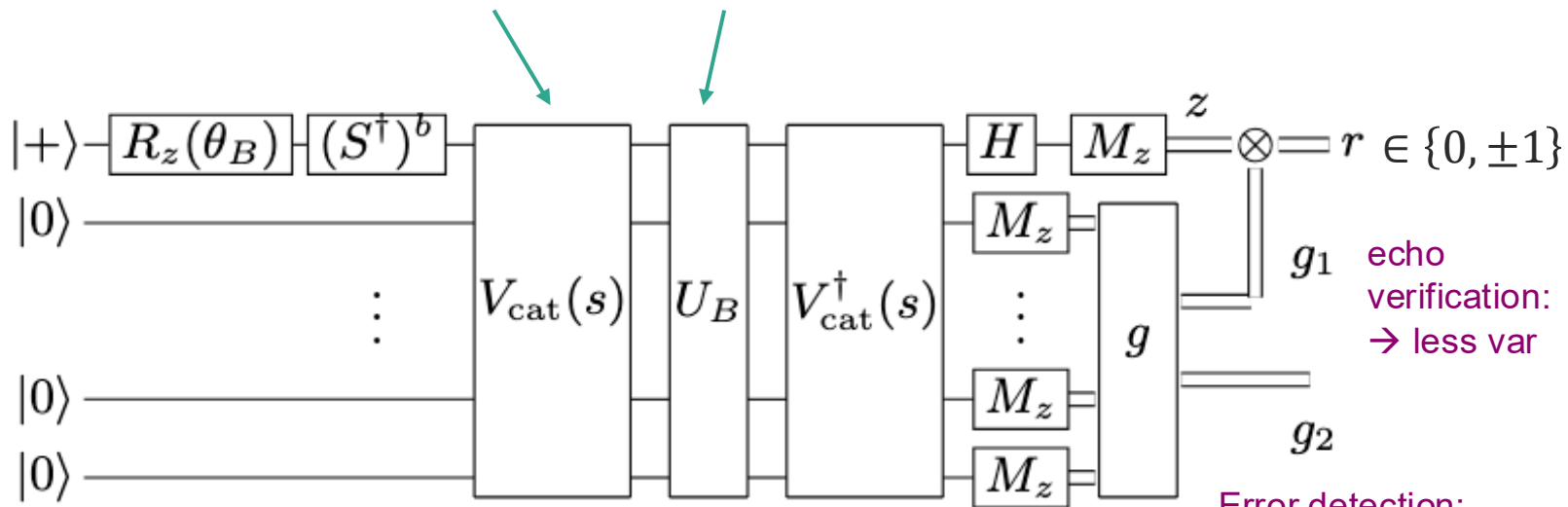
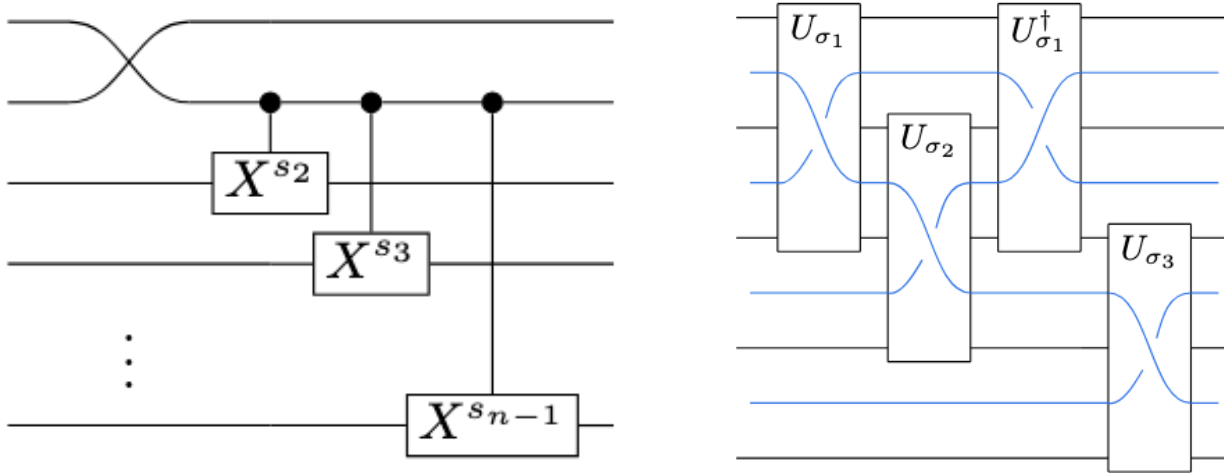
$$\text{Var}[O] = M^{-1} \left(\langle O^2 \rangle - \langle O \rangle^2 \right)$$

$$\text{Var}_{HT} = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = 1 - \text{Re}(\alpha)^2$$

$$\text{Var}_{CFEV} = \mathbb{E}[\tilde{Z}^2] - \mathbb{E}[\tilde{Z}]^2 = \frac{1}{2}(1 + |\alpha|^2) - \text{Re}(\alpha)^2$$

$$\text{Var}_{HT} - \text{Var}_{CFEV} = \frac{1}{2}(1 - |\alpha|^2) \geq 0 \text{ because } |\alpha| \leq 1$$

BQP version (Plat closure): estimate a weighted trace of U_B



Cat-state trick to remove ancilla:
→ fewer gates

Error detection:
→ discard non Fibo s

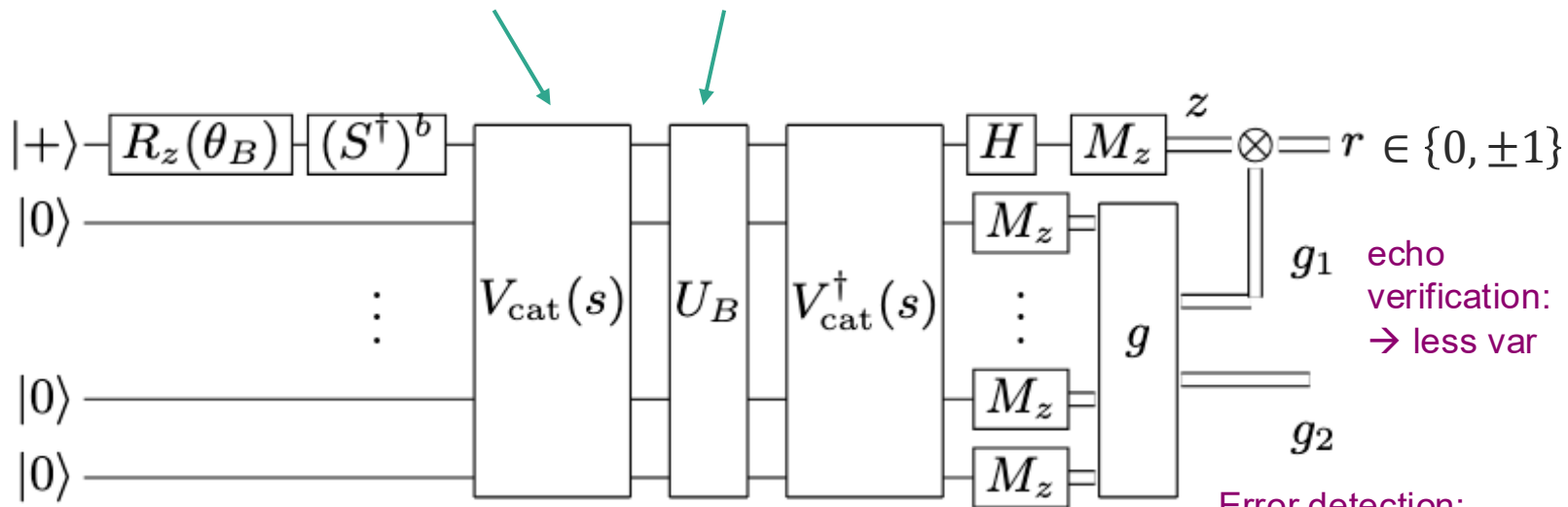
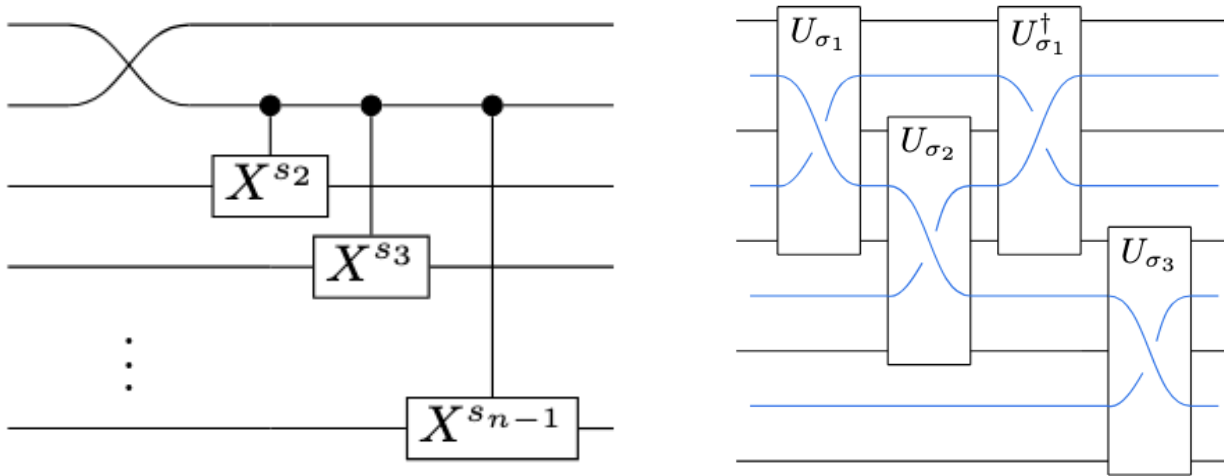
fix string $s = |010 \dots 010\rangle = \text{cups}$

$$V_{P(B)}(e^{i\frac{2\pi}{5}}) = (-e^{-i\frac{3\pi}{5}})^{3w_B} \phi^{\frac{n}{2}} (R + \epsilon).$$

$$\text{where } R = N^{-1} \sum_{i=1}^N r_i$$

and $\epsilon = \epsilon_{\text{shot}} + \epsilon_{\text{noise}}.$

DQC1 version (Markov closure): estimate a weighted trace of U_B



Cat-state trick to remove ancilla:
→ fewer gates

Error detection:
→ discard non Fibo s

sample Fibonacci strings:

$$s \sim p(s) = \begin{cases} \frac{\phi^{s_{n-1}}}{\phi^{n-1}} & \text{if } |s\rangle \in \mathcal{F}_n \\ 0 & \text{otherwise.} \end{cases}$$

$$V_{M(B)}(e^{i\frac{2\pi}{5}}) = (-e^{-i\frac{3\pi}{5}})^{3w_B} \phi^{n-2} (R + \epsilon)$$

$$\text{where } R = N^{-1} \sum_{i=1}^N r_i$$

$$\text{and } \epsilon = \epsilon_{\text{shot}} + \epsilon_{\text{noise}}.$$

End-to-end implementation of quantum algorithm:

<https://github.com/Quantinuum/jones-polynomial>

RESULTS FOR DQC1 VERSION

“Less quantum, more advantage”

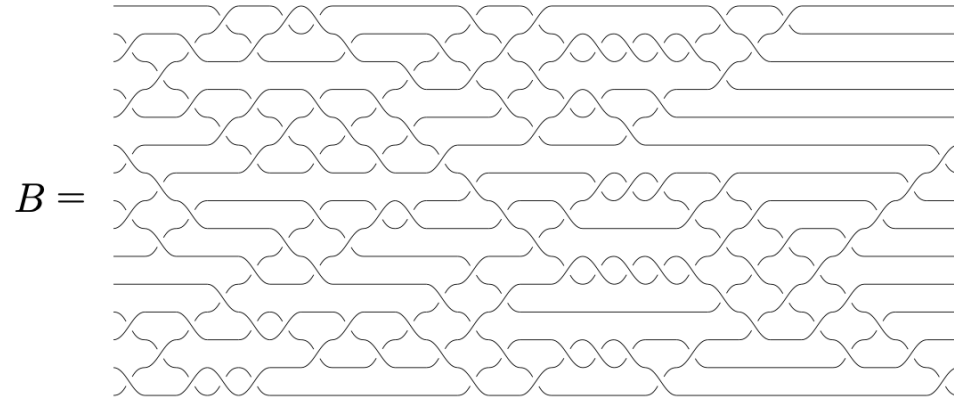
DQC1 is “less quantum” than BQP because it is believed that $DQC1 \subseteq BQP$.

However, there are no known efficient classical algorithms for DQC1 problems, so also promise for exponential advantage.

The DQC1 version of the problem is more challenging for classical algorithms than the BQP version. So there is potential for “more advantage”.

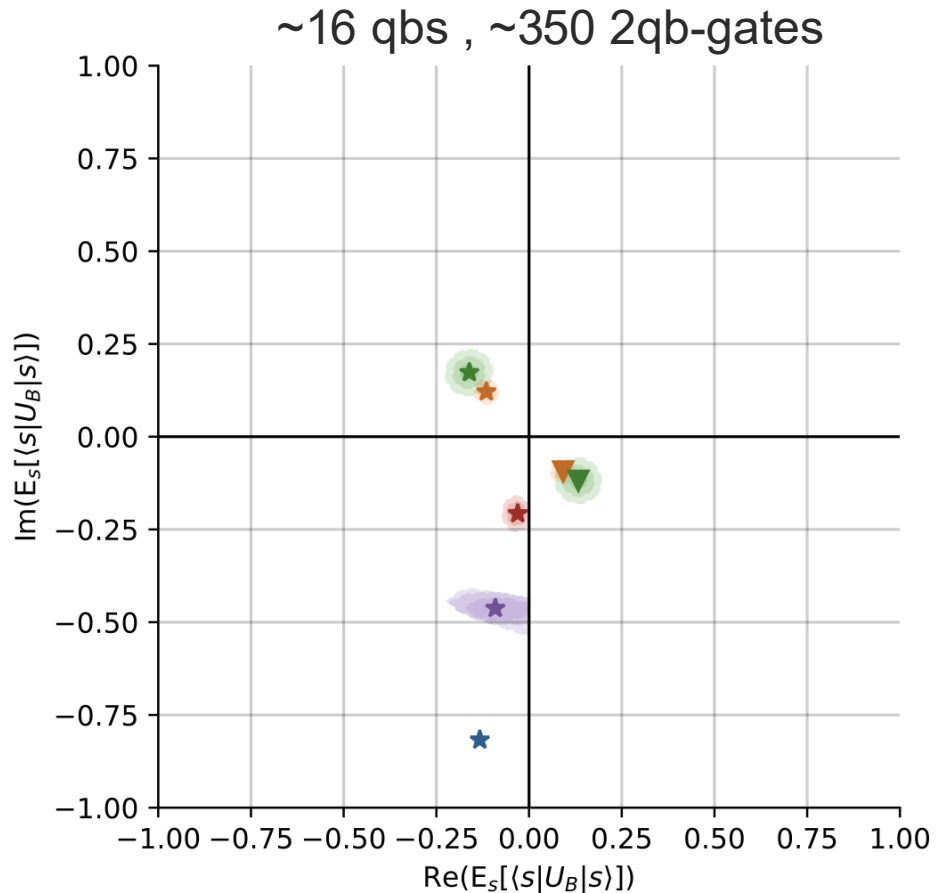
Demo Run on H2-2 (10e-3 2q err)

→ error detection and mitigation tricks work



$$M(B) \sim \text{[Diagram of two crossings]} \otimes \text{[Diagram of two crossings]}^{\otimes 3} \otimes \text{[Diagram of a single crossing]}^{\otimes 7}$$

- | | |
|--|---|
| ■ Ground Truth | ★ Original Braid |
| ■ Unmitigated | ▼ Conjugate Braid |
| ■ + Error Detection | ■ 3σ Confidence Interval |
| ■ + Conjugate Trick | |
| ■ + Shot-Level Conj. Trick | |



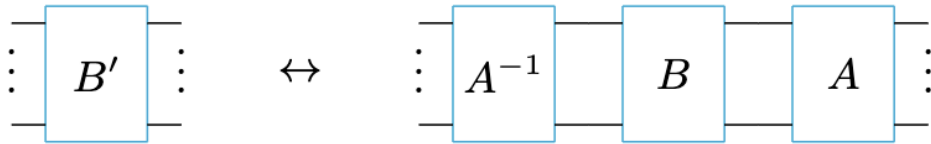
Conjugate trick to mitigate memory error
(B^* is braid obtained by flipping all crossings,
so need to run 4 circuits instead of 2):

$$R = \pm \frac{|e^{i\theta} R + e^{i\theta} R^*|}{2} \pm i \frac{|e^{i\theta} R - e^{i\theta} R^*|}{2}$$

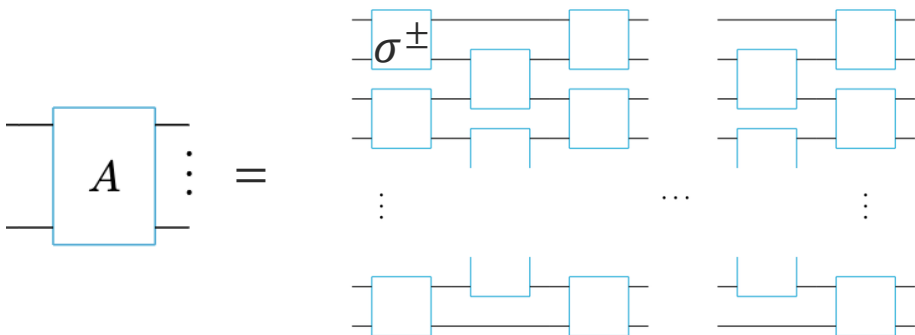
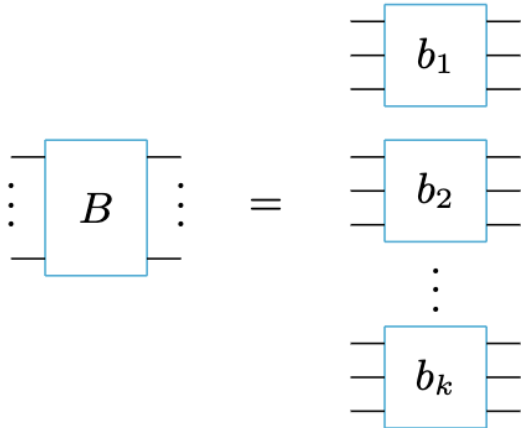
Benchmark to characterize gate-noise ϵ_{noise}

*Efficiently verifiable
Problem-tailored*

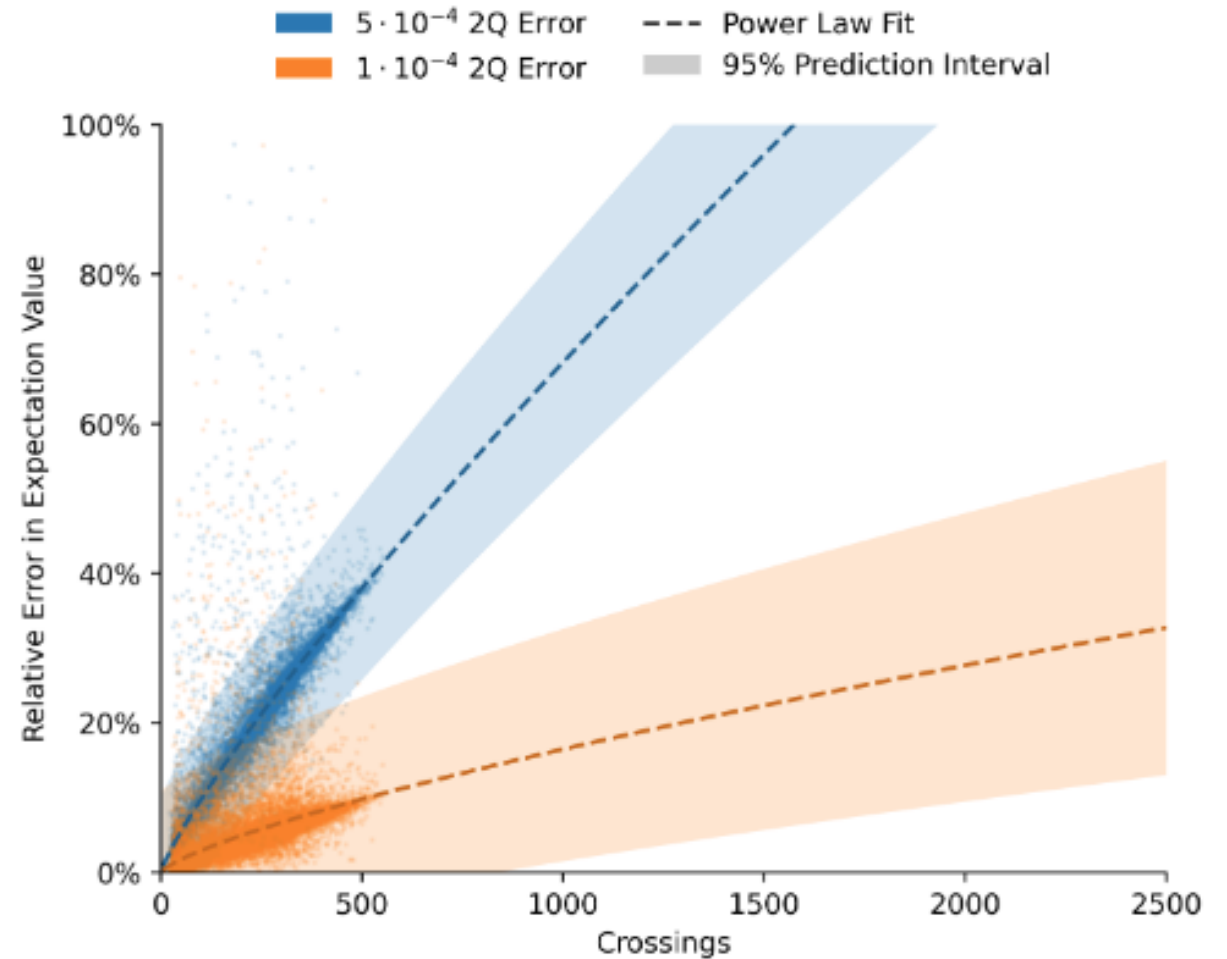
Augment braids with **known Jones value**,
create set of topologically equivalent braids



where

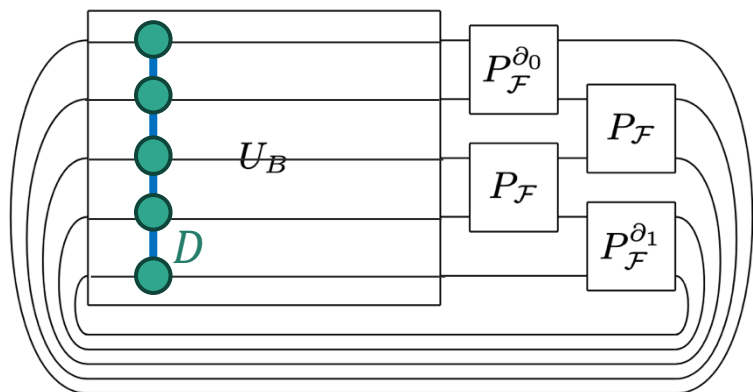


Extrapolation from **noisy simulation** (depolarizing noise)



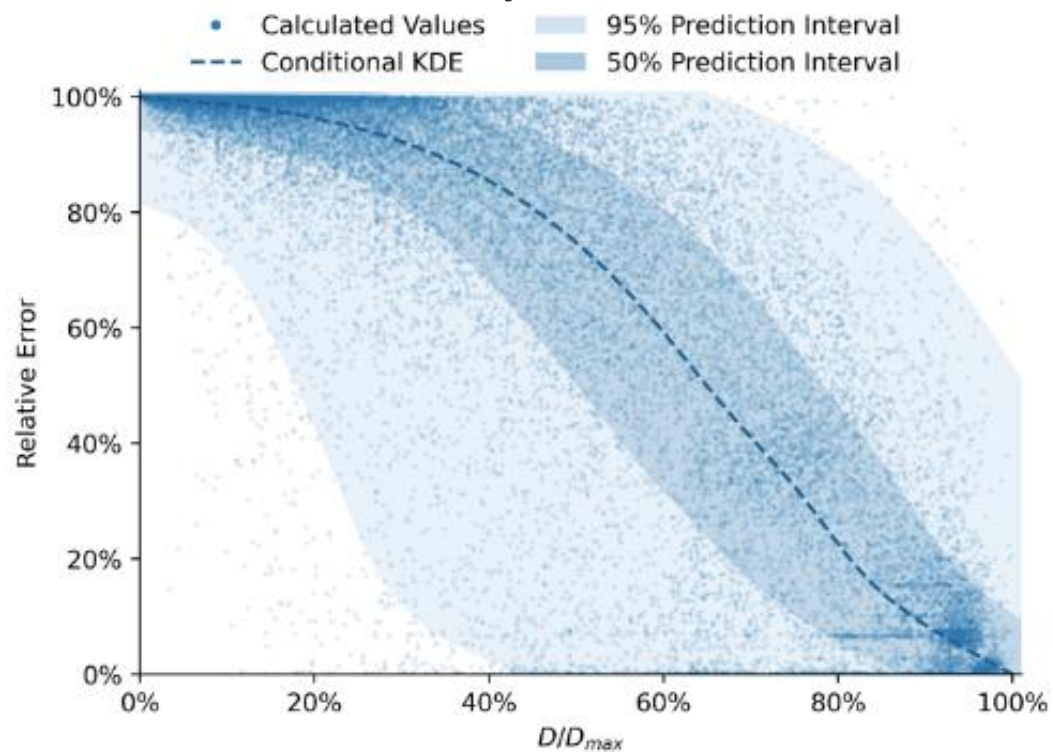
** On a big fast quantum computer
it would be interpolation.*

Best (we think) Classical Algo:

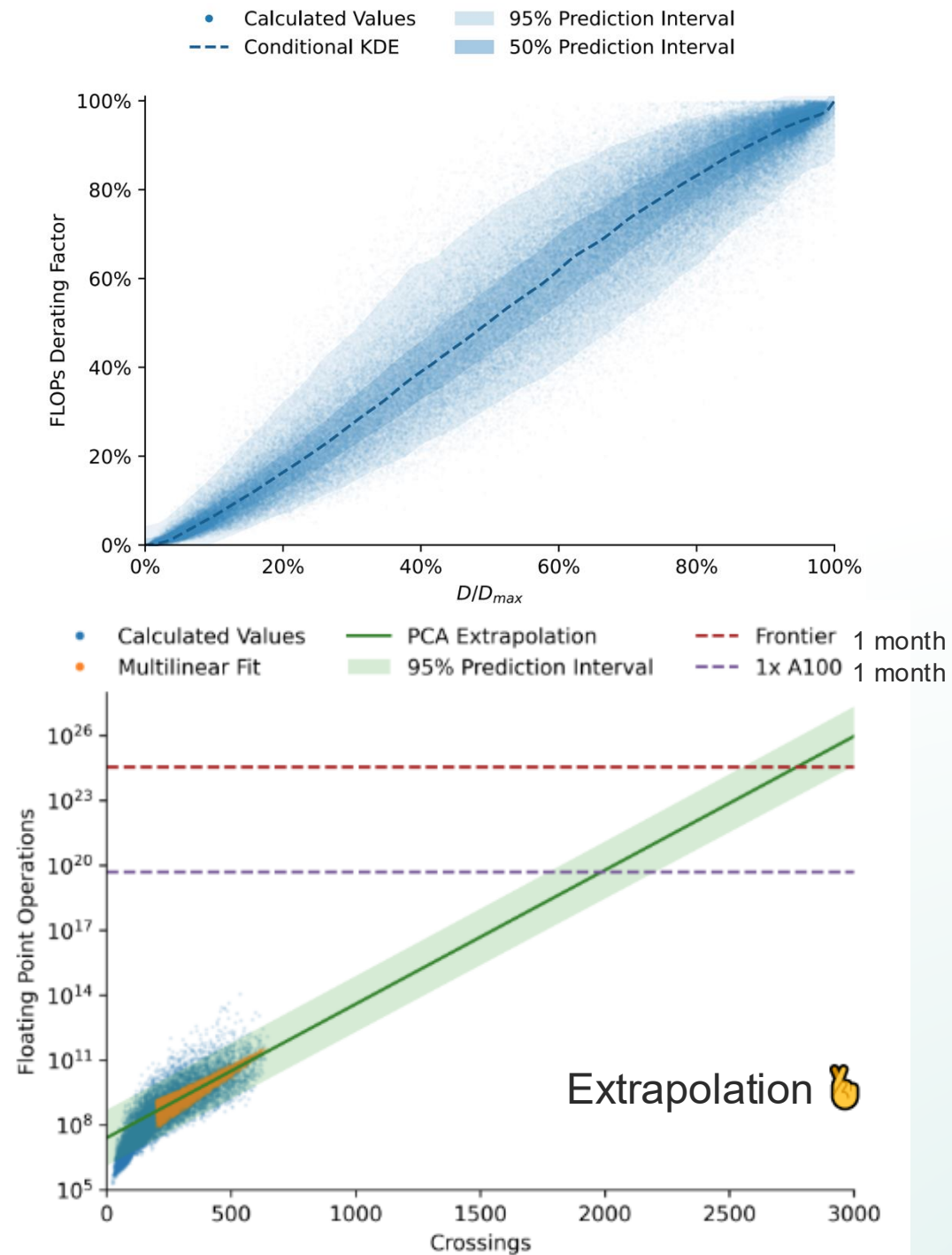


MPO with
Fibonacci projectors
and trace

memory vs error



operations vs braid size



Locating Quantum Advantage

quadvantage = shorter quantum time-to-solution for given additive error

Quadvantage hunting logic:

Given a QC with [gate-fidelity, layer-time]

Given a CC with [RAM, clock speed]

→ how long does CC need to achieve same error?

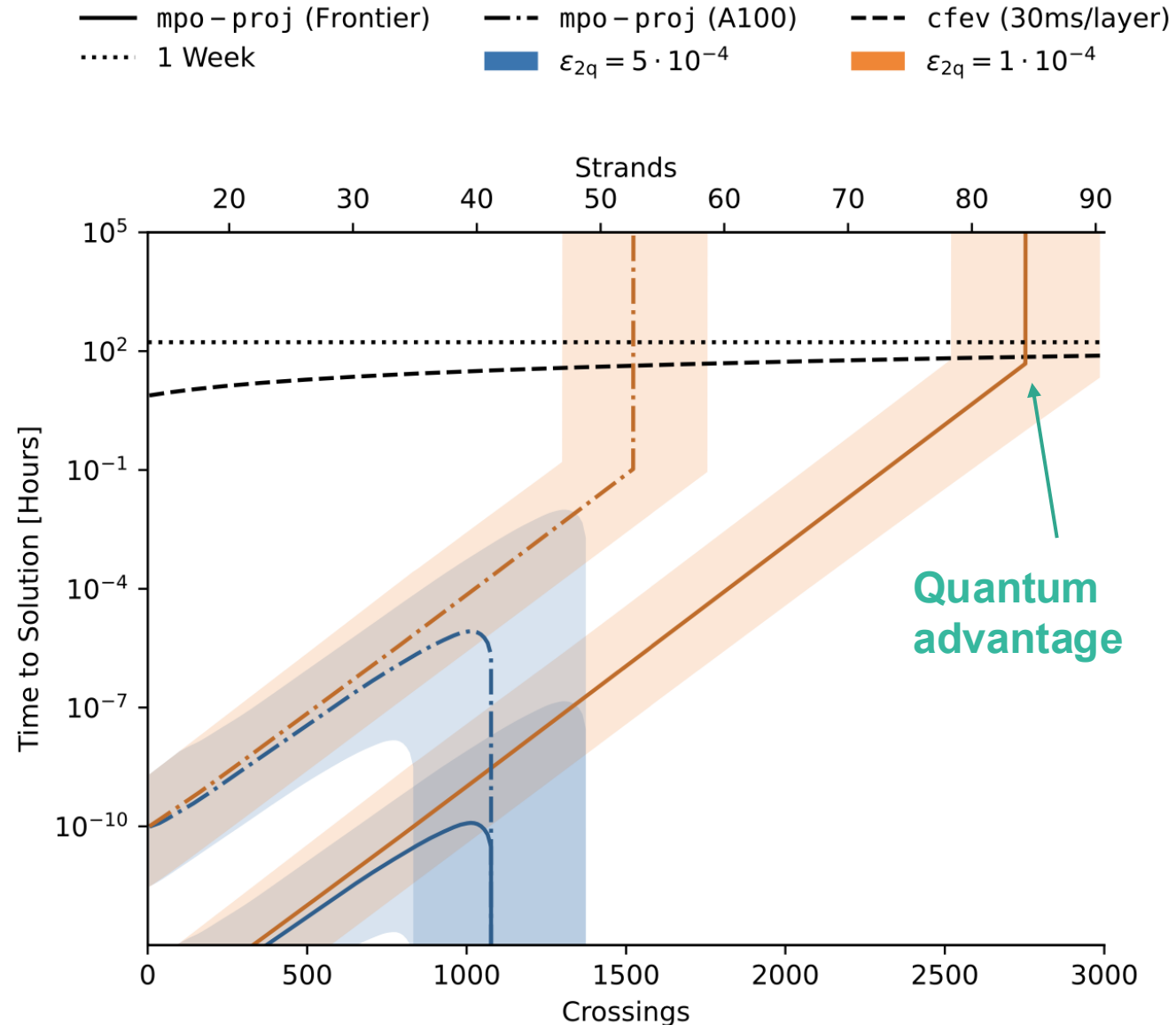
DQC 1-complete

→ **queasy** braids **must** exist

Quantum advantage minimum instance size:

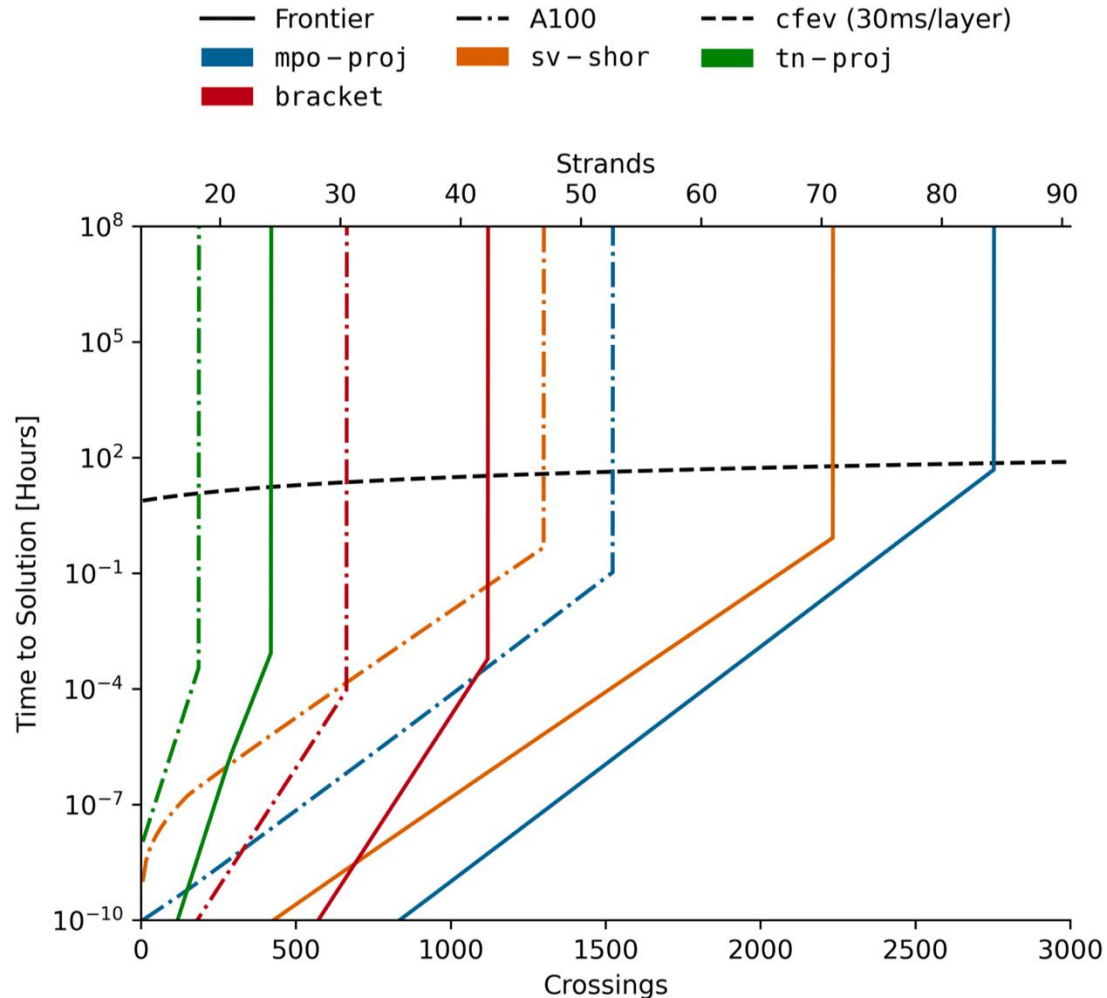
~85 qubits/strands

~2.8k crossings = 8.4k 2-qubit gates



Locating Quantum Advantage

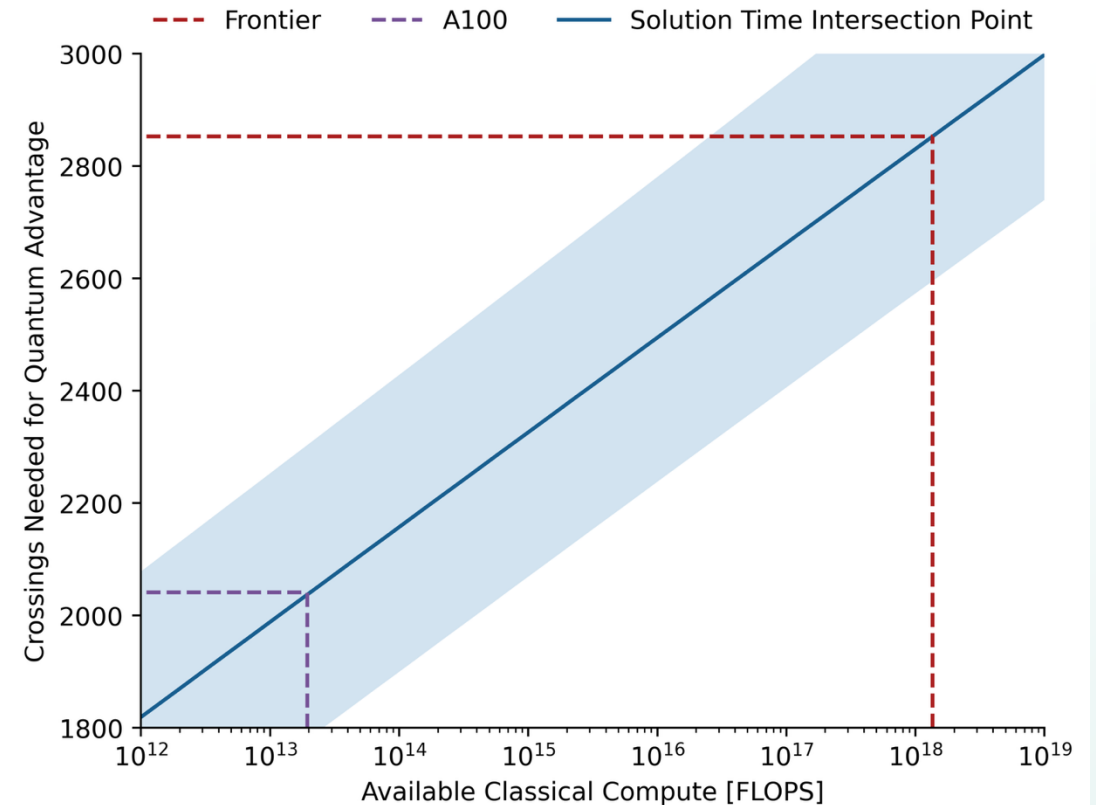
Btw, we tried a lot of classical algorithms and mpo-proj is best:



Assuming:

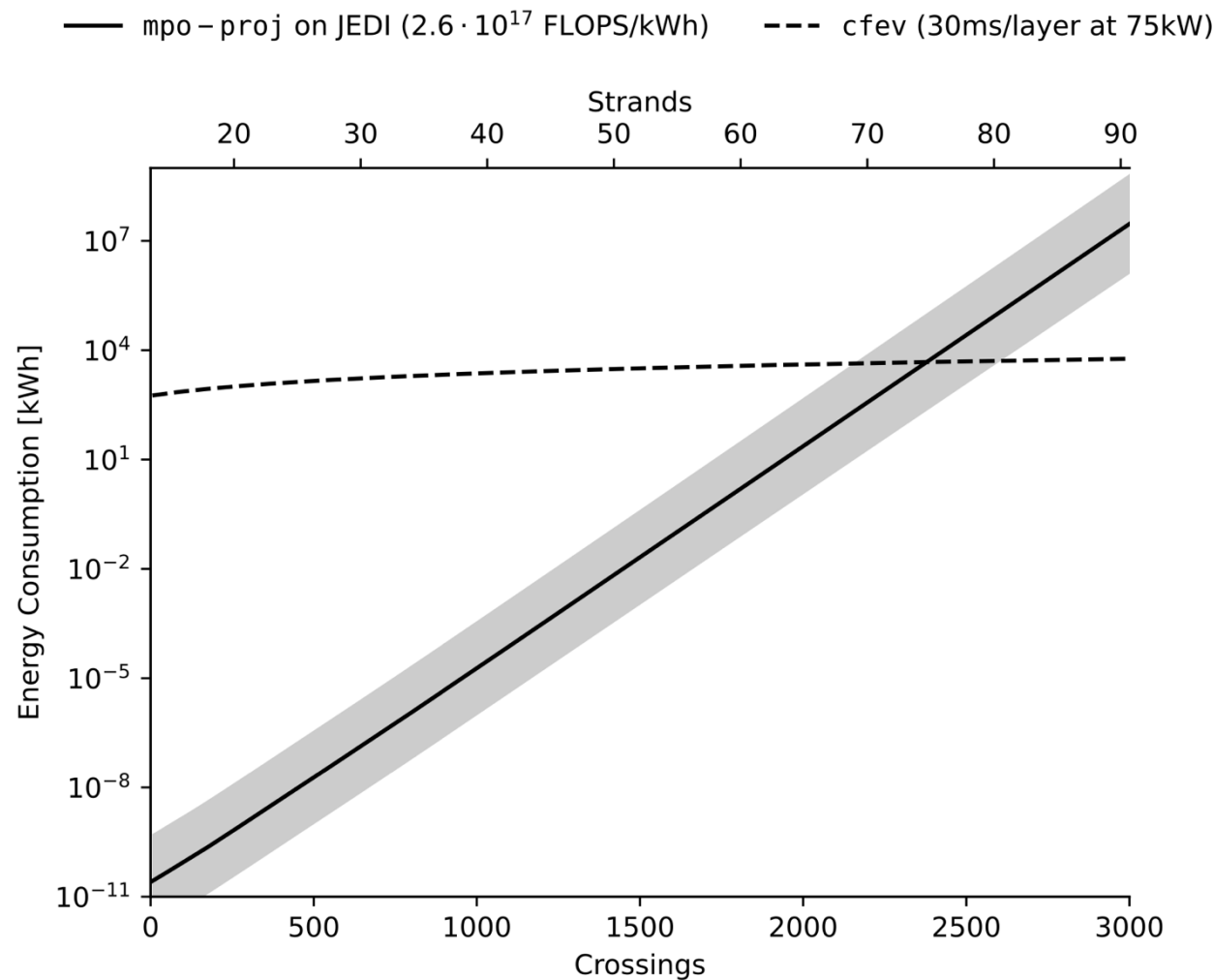
- classical compute in x FLOPS/s
- quantum computer with gate error $1e-4$ and 30ms/layer

→ what problem size y is required for quadvantage?

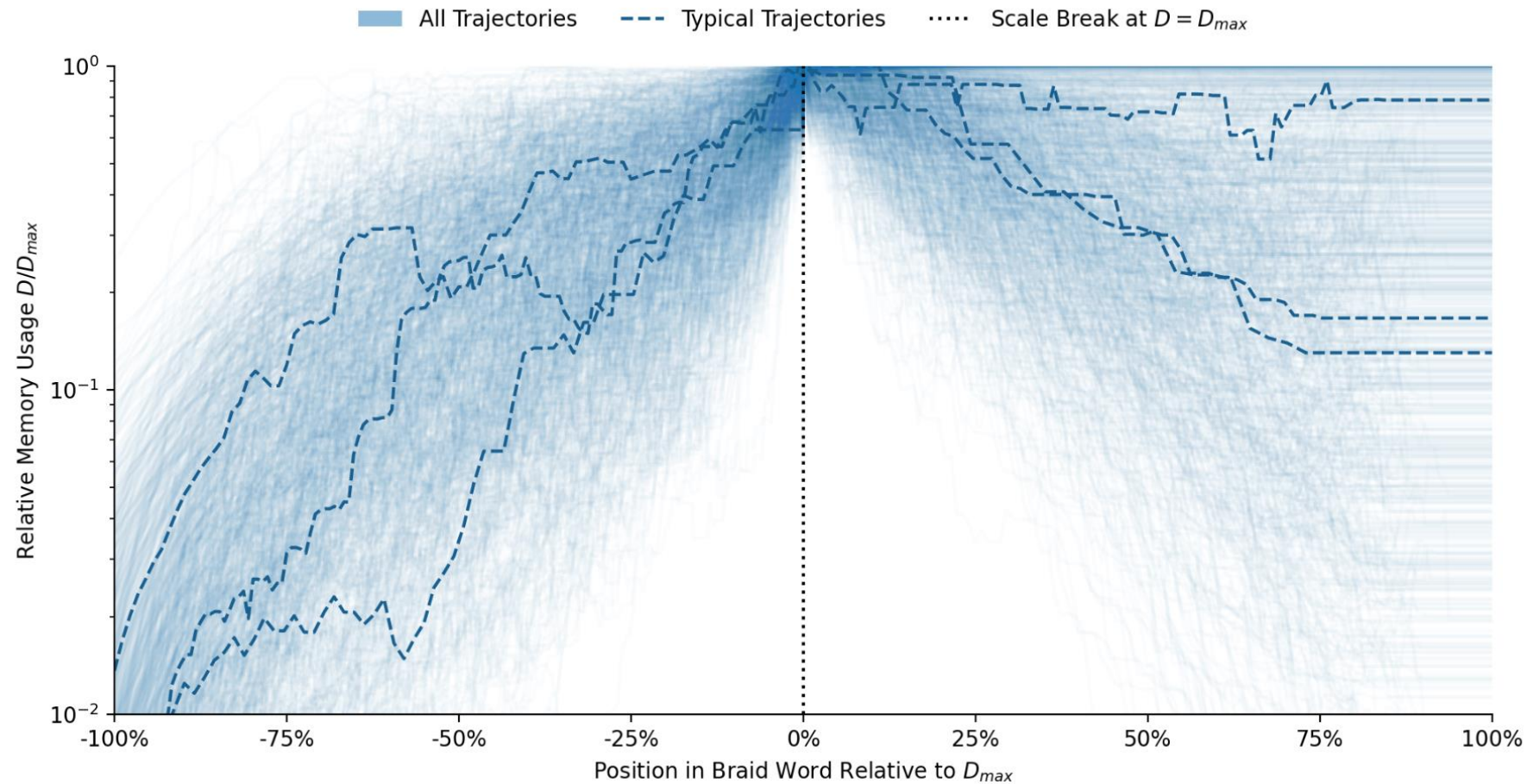


Energy Consumption Advantage

quadvantage = lower energy consumption given additive error



A limitation



The MPO can “see” the conjugate structure of the braids in the benchmark set

We have identified the minimum instance **sizes**.

But for a random braid of this size, the classical algorithm is easy: “print 0”. And will be correct with high prop.

Random Fibo braids scramble fast!

They are like random universal circuits, so all amplitudes are almost the same (up to exp small fluctuations).

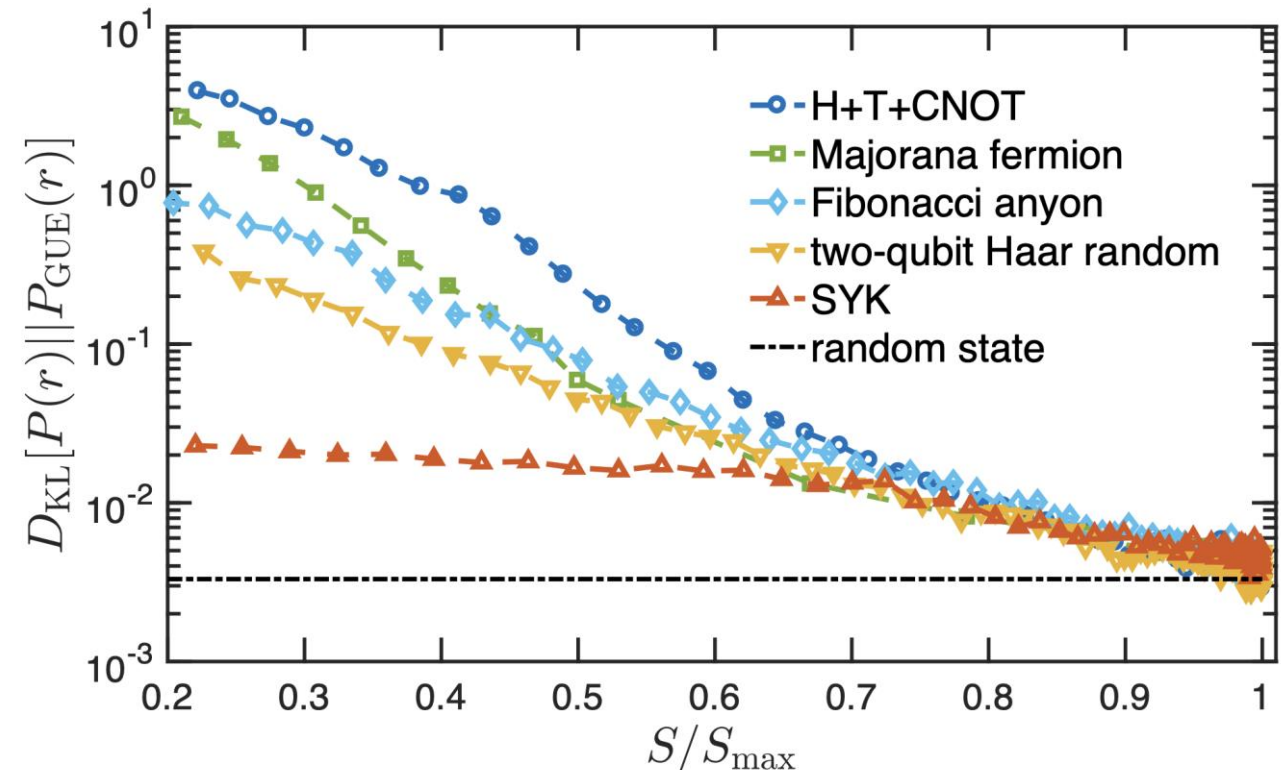
λ : entanglement spectrum of state
during random Fibo evo

$$r_k = \frac{\lambda_{k-1} - \lambda_k}{\lambda_k - \lambda_{k+1}}$$

Gap ratios approach GUE : quantum chaos

$$P_{\text{WD}}(r) = \frac{1}{Z} \frac{(r + r^2)^\beta}{(1 + r + r^2)^{1+3\beta/2}},$$

with $Z = 4\pi/81\sqrt{3}$ and $\beta = 2$ for GUE distribution.

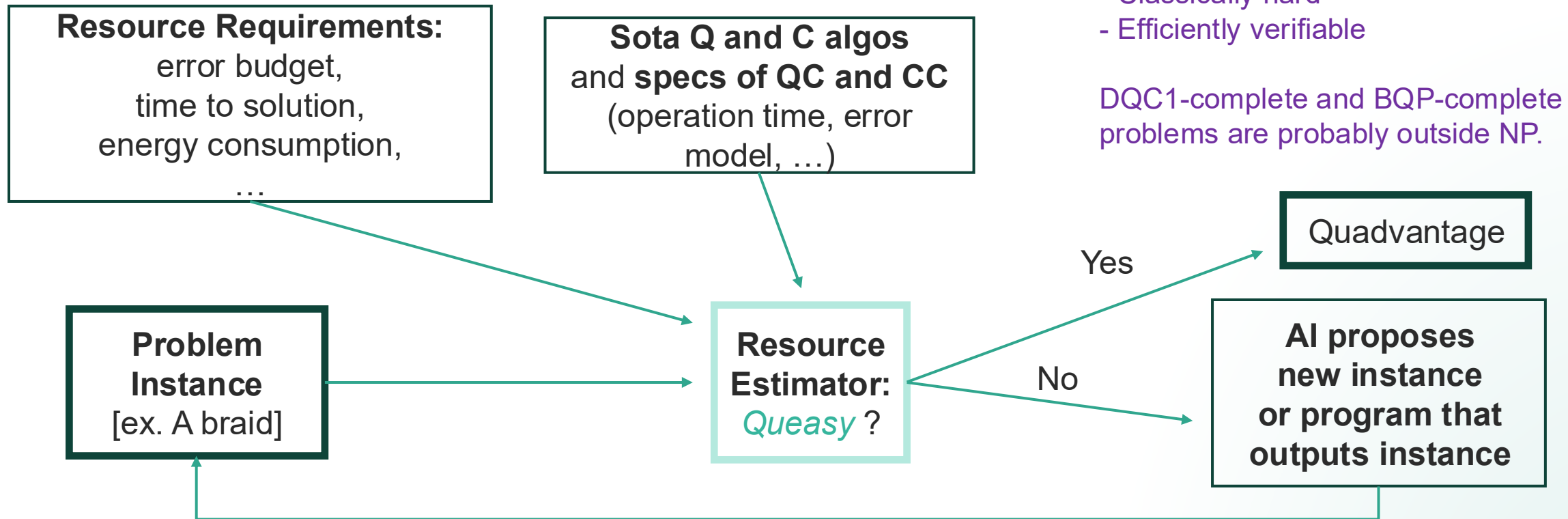


Queasy braids?

We can construct queasy (technical term [2510.01953]) braids by reduction from another BQP problem.

Example: compile braid $U = \exp -iHt$ (t can be a Trotter step and then repeat for longer times), compile braid $V = X$, make braid $UVU^\dagger$ and Plat-close. You get a braid that evaluates to a non-vanishing physical observable, for example H being a 2D Ising model.

Alternatively: AI-driven Quadvantage Hunting!



WHEN IS JONES *EASY* ?

A unified ZX
calculus perspective

ZX diagrams

Tensor networks composed of two tensors (“spiders”):

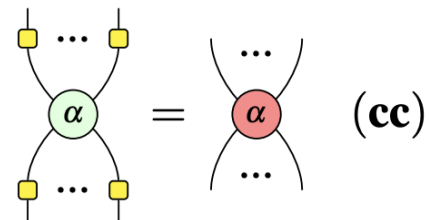
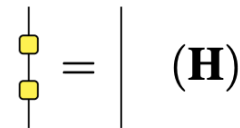
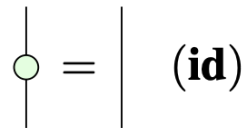
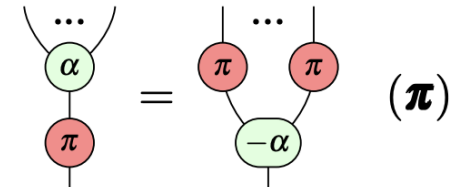
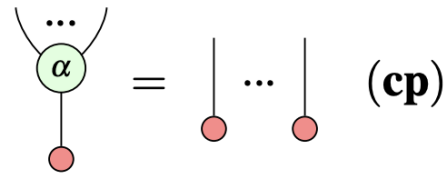
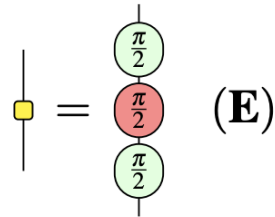
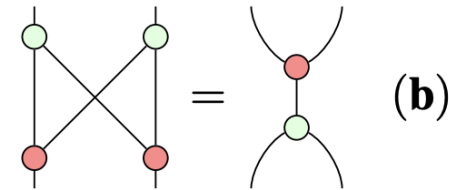
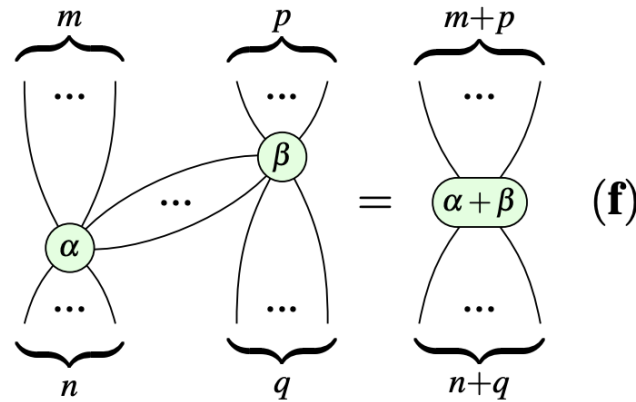
Z-spider : **Copy in Z basis,
and apply phase.**

$$\begin{array}{c} \cdots \\ \diagup \quad \diagdown \\ \textcircled{\alpha} \\ \diagdown \quad \diagup \\ \cdots \end{array} = |0 \dots 0\rangle \langle 0 \dots 0| + e^{i\alpha} |1 \dots 1\rangle \langle 1 \dots 1|, \quad \alpha \in [0, 2\pi)$$

X-spider : **Copy in X basis,
and apply phase.** $|\pm\rangle = |0\rangle \pm |1\rangle$

$$\begin{array}{c} \cdots \\ \diagup \quad \diagdown \\ \textcircled{\alpha} \\ \diagdown \quad \diagup \\ \cdots \end{array} = |+\dots+\rangle \langle +\dots+| + e^{i\alpha} |-\dots-\rangle \langle -\dots-|, \quad \alpha \in [0, 2\pi)$$

Qubit ZX Rewrite Rules

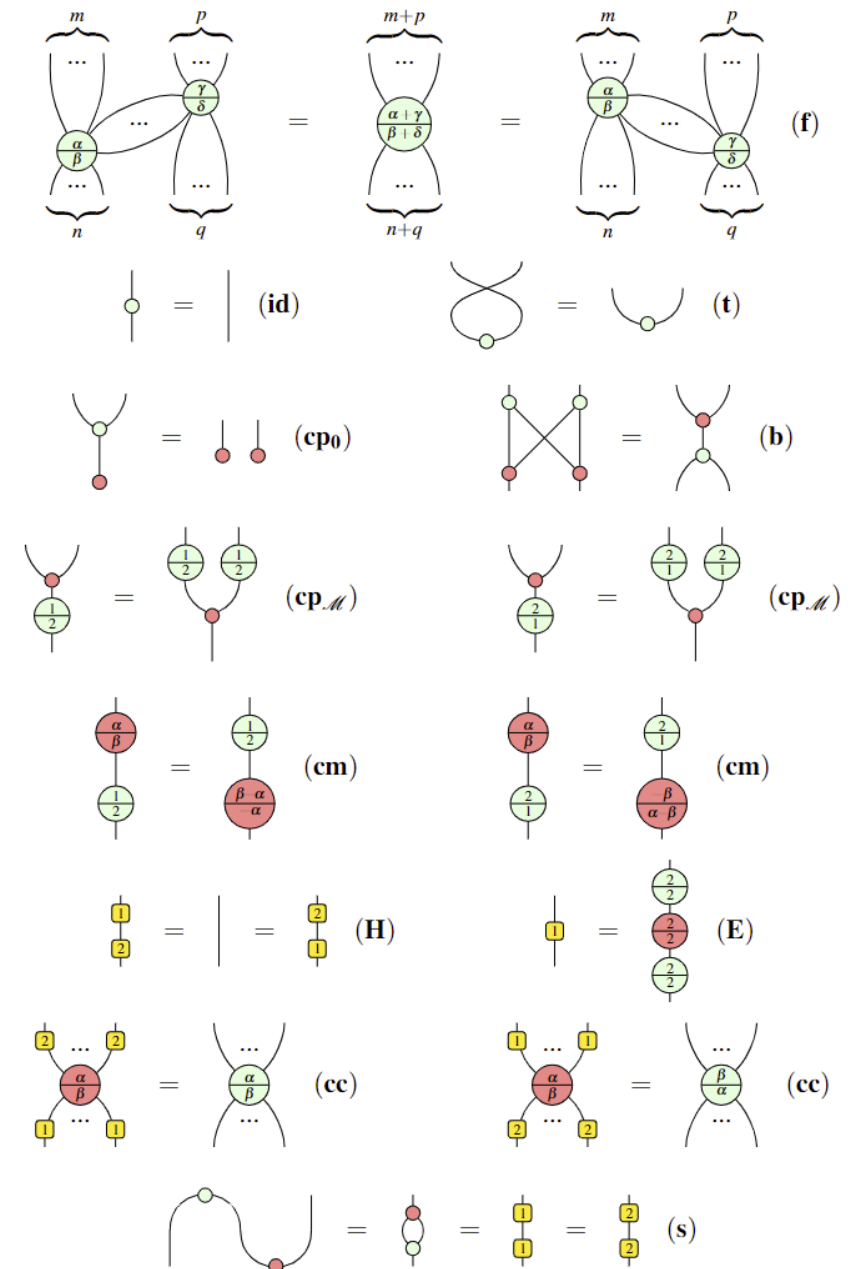


Qutrit ZX Rewrite Rules

basis $\{|0\rangle, |1\rangle, |2\rangle\}$

ZX also works for qudits of
of any prime dimension.

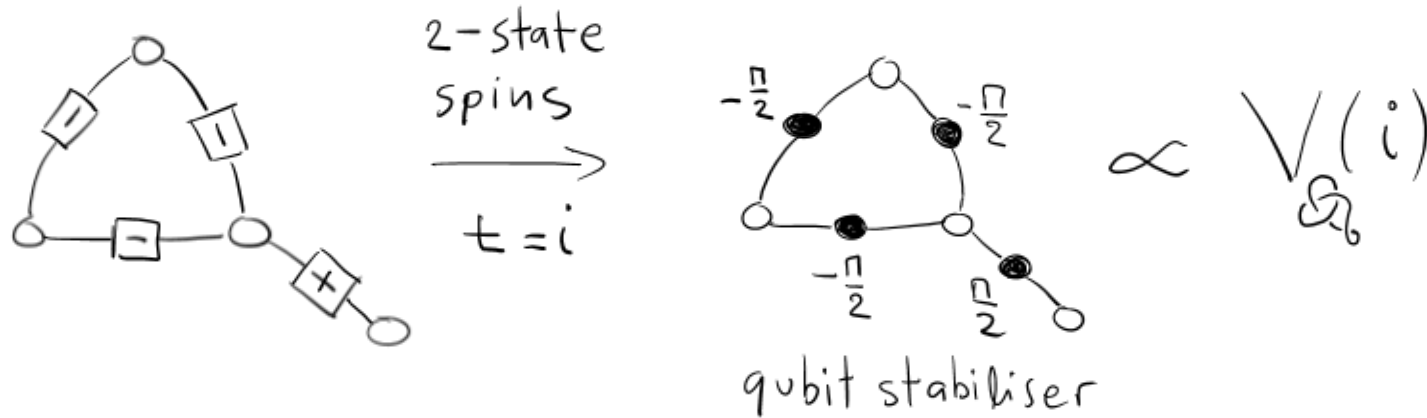
Check work of Quanlong
Wang.



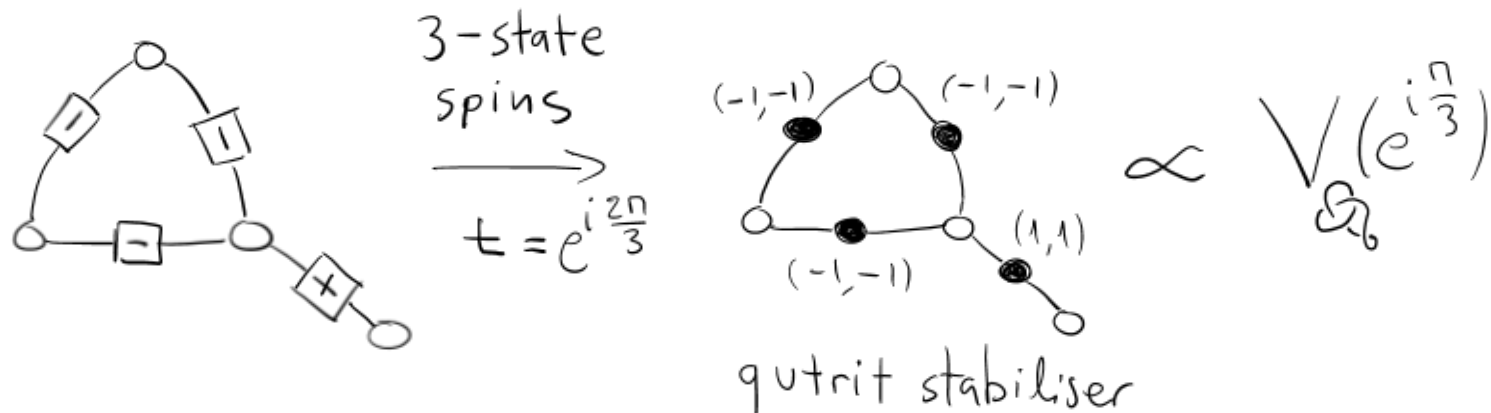
At $t = i$ (q=2) and $t = e^{i\frac{\pi}{3}}$ (q=3):

the problem is equivalent to rewriting stabilizer ZX diagrams
which are easy to simulate quantum computations
qubit and qutrit Clifford circuits!

Jones via Potts at lattice roots is stabilizer ZX



For example, for
Fibonacci $q=\varphi+1$



Qubit Stabilizer ZX Simplification

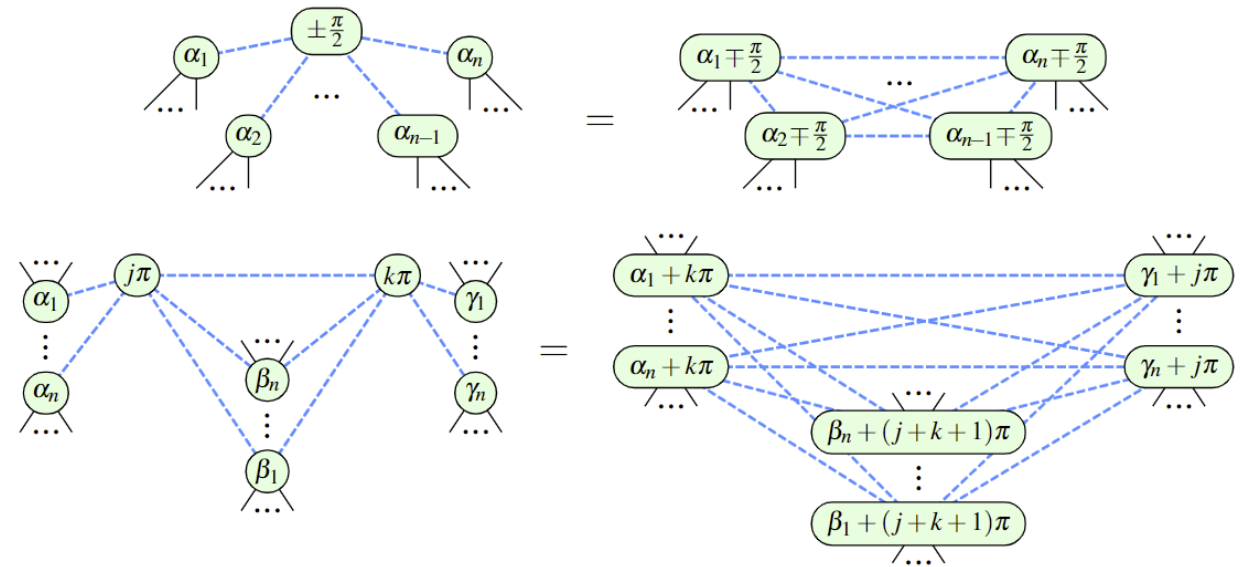
Graphical Gottesman-Knill

Computing amplitudes is, in general, hard!

Except when the phases
of the spiders are multiples of $\frac{\pi}{2}$:
we call such diagrams Stabiliser diagrams.

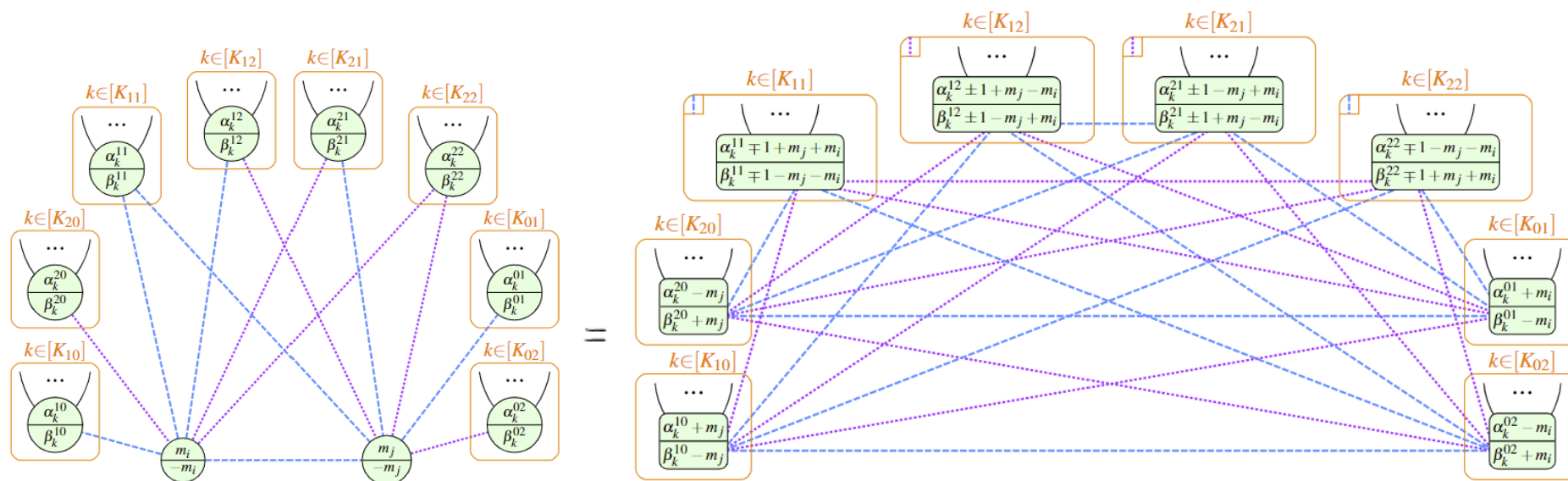
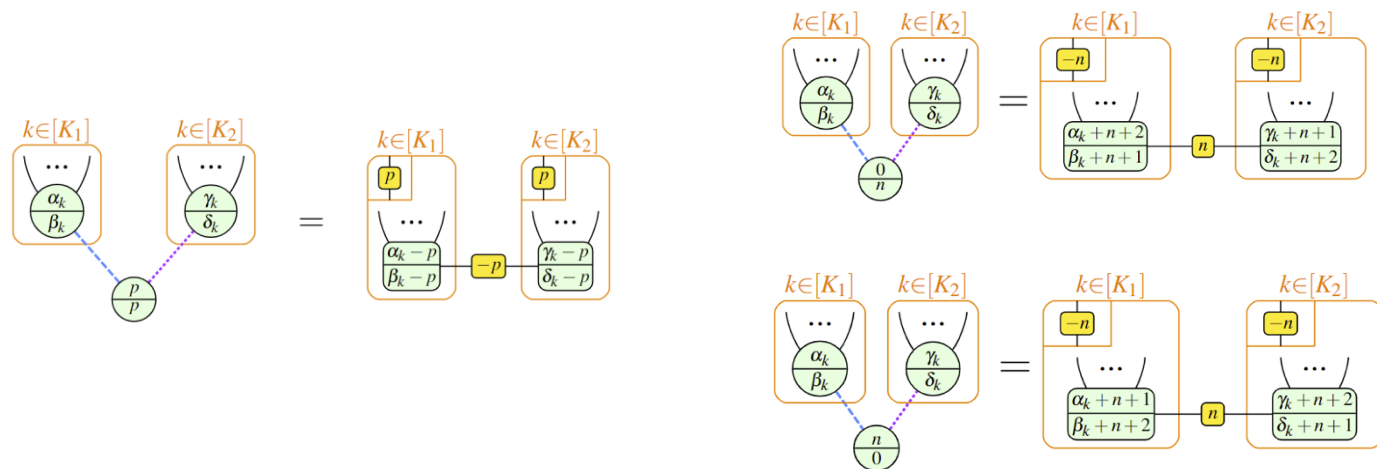
Stabiliser diagrams are **easy to simplify**.

spider-killing rules



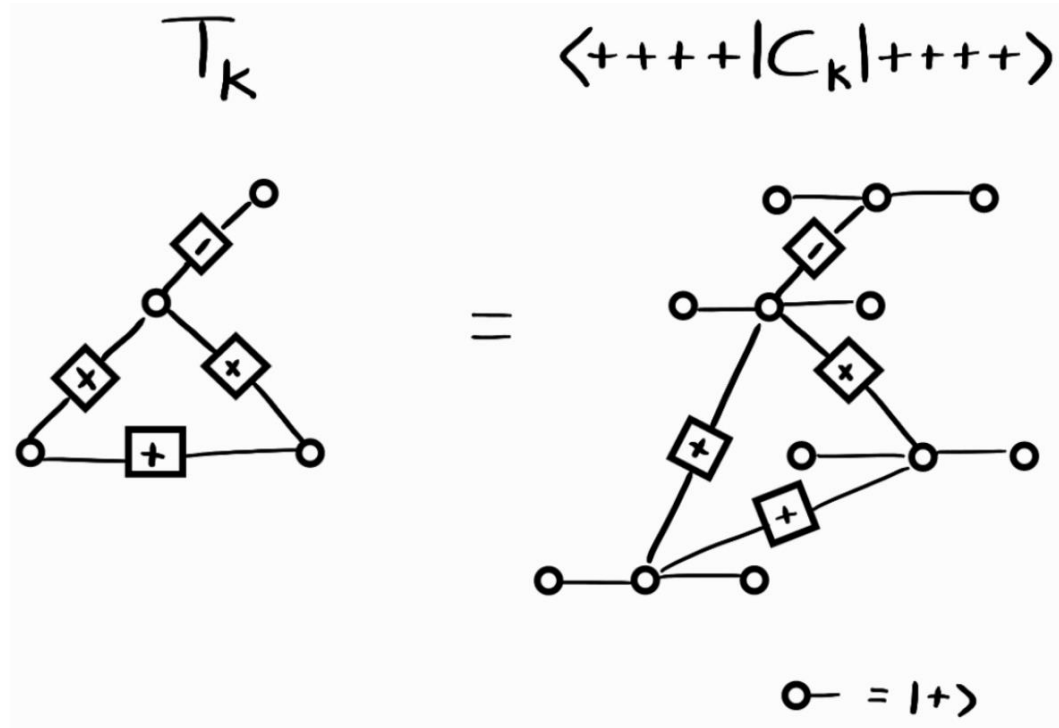
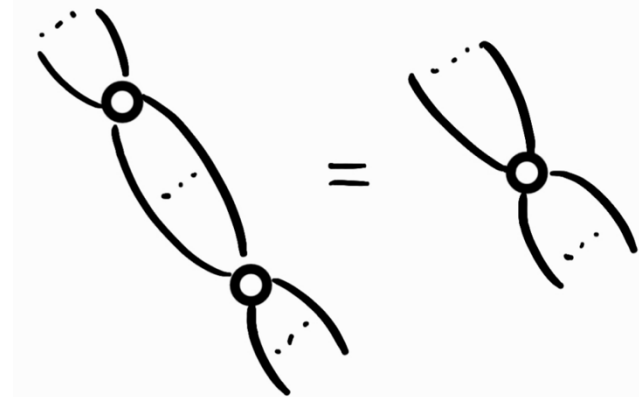
Qutrit Stabilizer ZX Simplification

Qutrit stabiliser diagrams are those where spiders carry phases that are multiples of $\frac{\pi}{3}$. They can be simplified **efficiently**.



spider-killing
rules

Spider fission $\rightarrow$ Clifford IQP amplitude

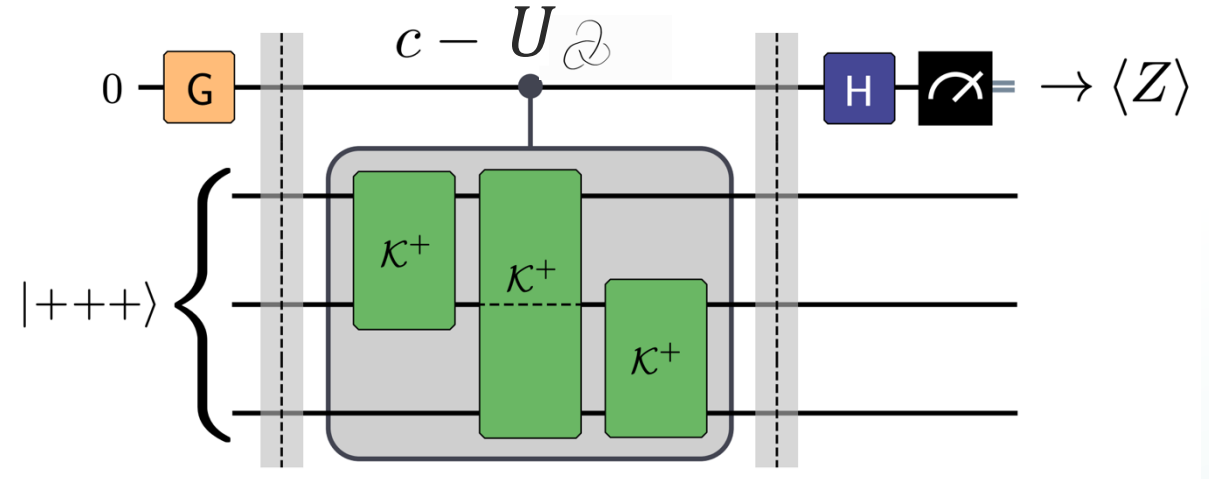
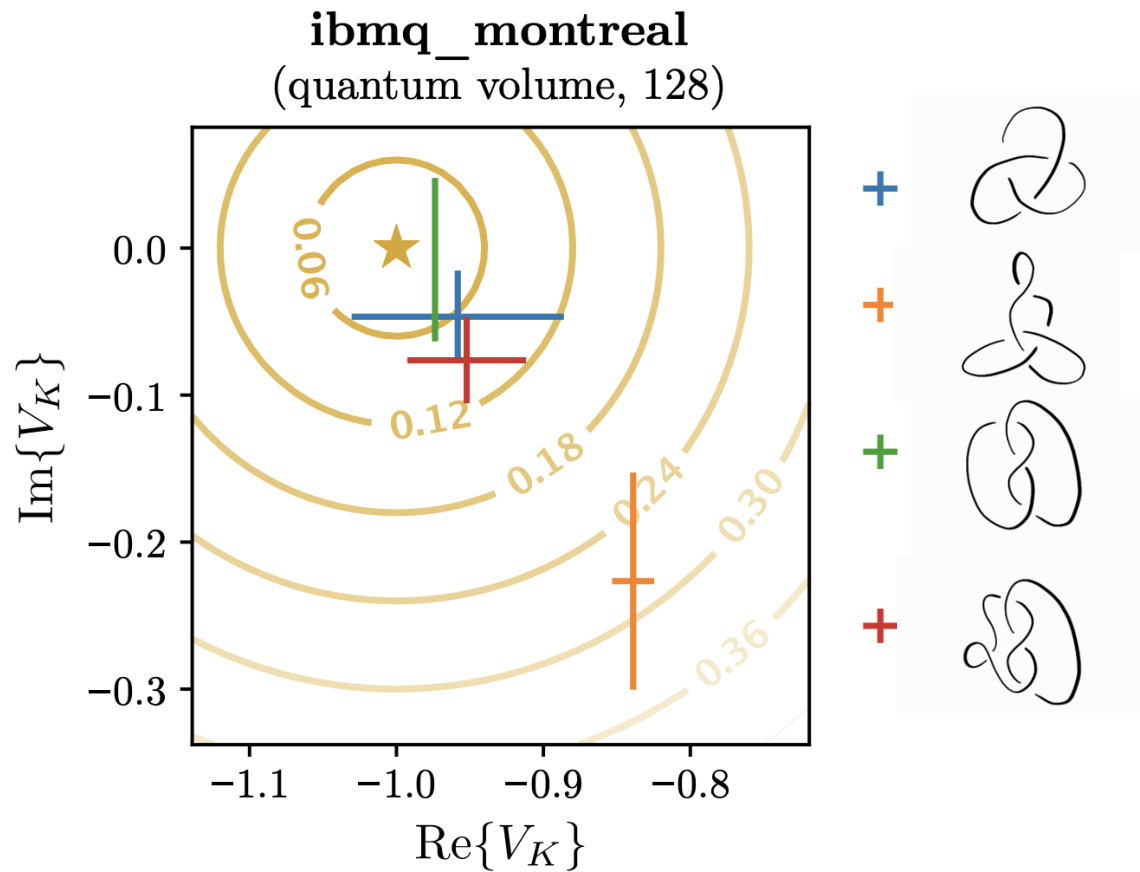


$$K^{\pm} = \begin{array}{c} \epsilon_i - \epsilon_k \\ \boxed{\pm} \\ \epsilon_j - \epsilon_l \end{array} = 1 - (1 + t(q)) \delta_{\epsilon_i \epsilon_j} \delta_{\epsilon_i \epsilon_k} \delta_{\epsilon_j \epsilon_l}$$

$$= \begin{cases} -t(q), & \epsilon_i = \epsilon_j = \epsilon_k = \epsilon_l \\ 1, & \epsilon_i = \epsilon_k \text{ and } \epsilon_j = \epsilon_l \\ 0, & \text{otherwise} \end{cases}$$

IBM demos (in 2021)

Evaluation at $t = i$ (4th root of unity) and using ZNE to mitigate errors.



$$\mathcal{K}^{\pm} = \text{diag}(\pm i, 1, 1, \pm i)$$

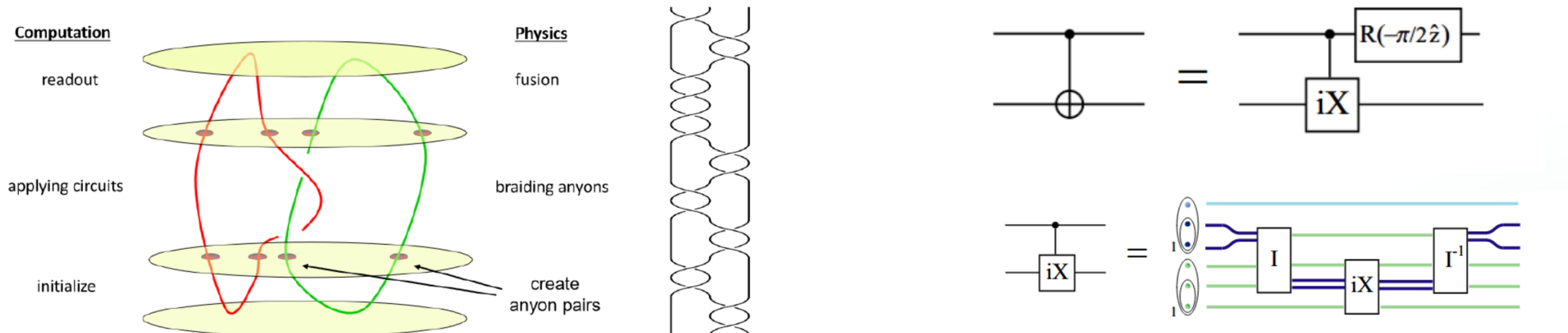
BONUS MISC FUN FACTS

good prompts for your mind

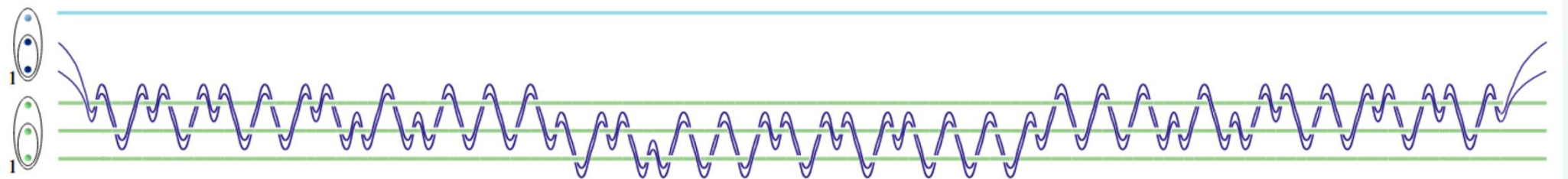
TQFT Black Magik and TQC

In 1989, Witten showed that if you compute the vacuum expectation value of a Wilson loop (the path of a charged particle in spacetime) within Chern-Simons theory, the result is exactly the Jones polynomial of that loop. Chern-Simons theory is the standard effective low-energy field theory used to describe the topological order of the Fractional Quantum Hall Effect (FQHE).

This TQFT framework directly underpins non-Abelian anyons in topological quantum computation (TQC). Braiding these anyons executes unitary gates, and measuring the fusion outcomes corresponds precisely to the BQP-complete problem of evaluating the Jones polynomial.



Adapted from [Rowell Wang 18](#), [Rouabah 20](#)



[Hormozi, Zikos, Bonesteel & Simon (2007)]

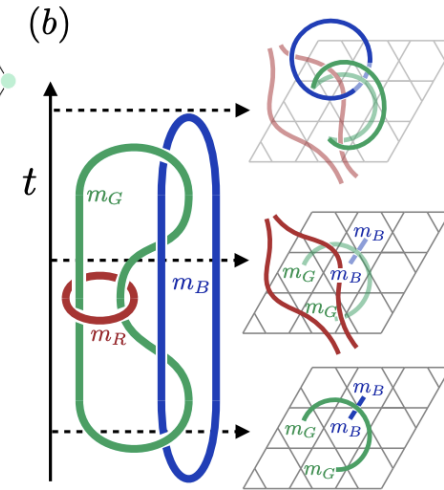
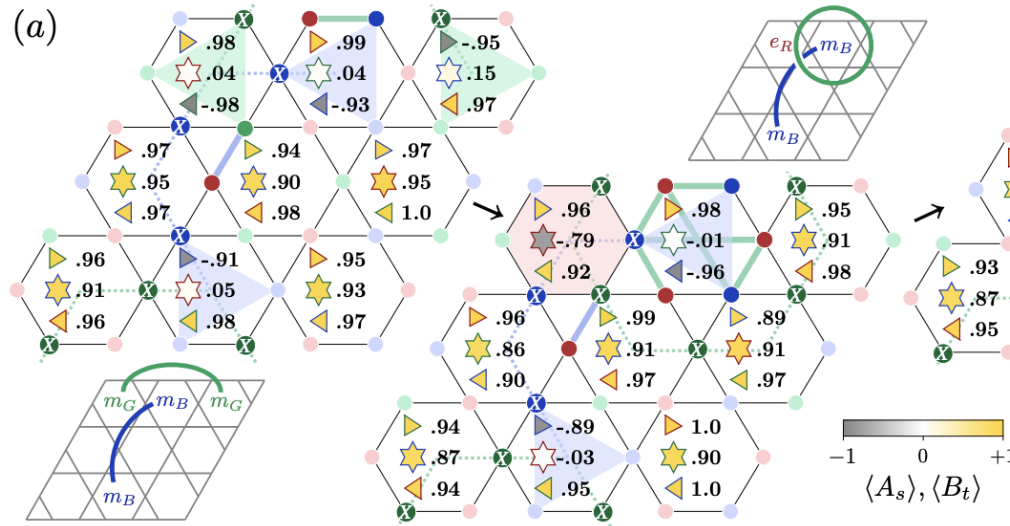
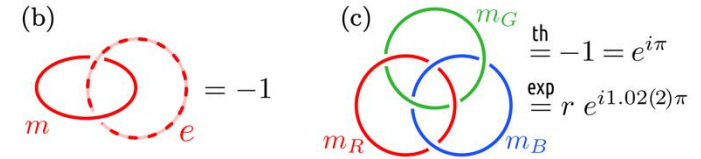
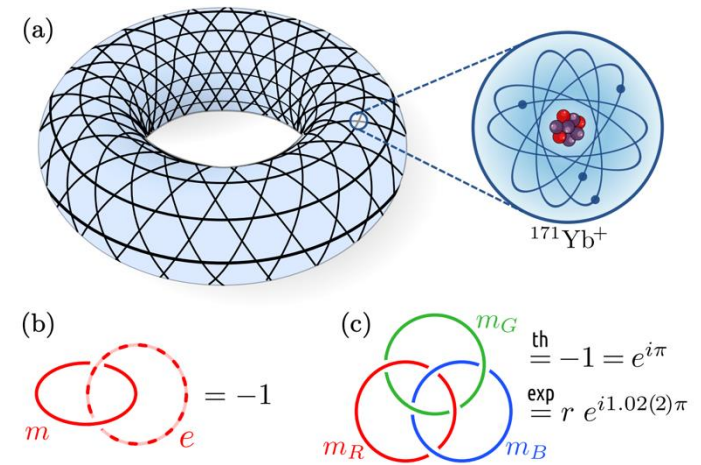
Quantum simulation of non-Abelian Anyons in stabilizer codes

<https://www.quantamagazine.org/physicists-create-elusive-particles-that-remember-their-pasts-20230509/>

Non-Abelian topological order and anyons on a trapped-ion processor

Mohsin Iqbal, Nathanan Tantivasadakarn, Ruben Verresen, Sara L. Campbell, Joan M. Dreiling, Caroline Figgatt, John P. Gaebler, Jacob Johansen, Michael Mills, Steven A. Moses, Juan M. Pino, Anthony Ransford, Mary Rowe, Peter Siegfried, Russell P. Stutz, Michael Foss-Feig, Ashvin Vishwanath & Henrik Dreyer ✉

Nature **626**, 505–511 (2024) | [Cite this article](#)



Similar experiments:

Non-Abelian braiding of graph vertices in a superconducting processor

[Google Quantum AI and Collaborators](#)

Nature **618**, 264–269 (2023) | [Cite this article](#)

Digital simulation of projective non-Abelian anyons with 68 superconducting qubits

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QTDA and Khovanov homology

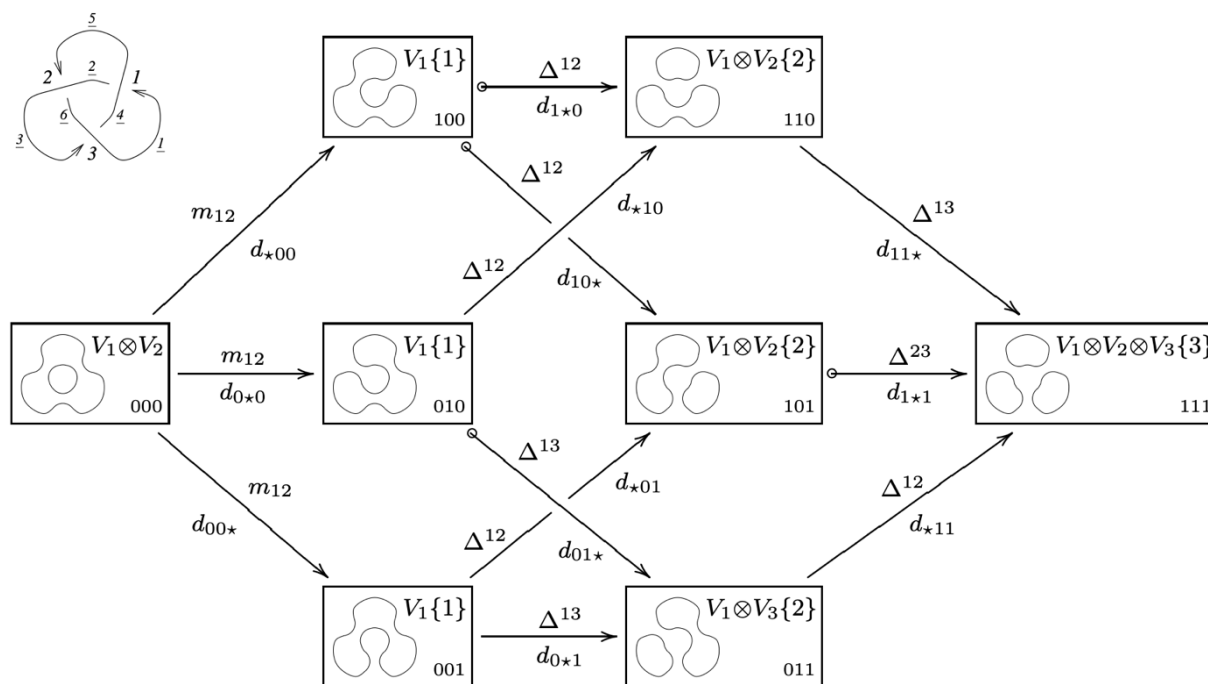
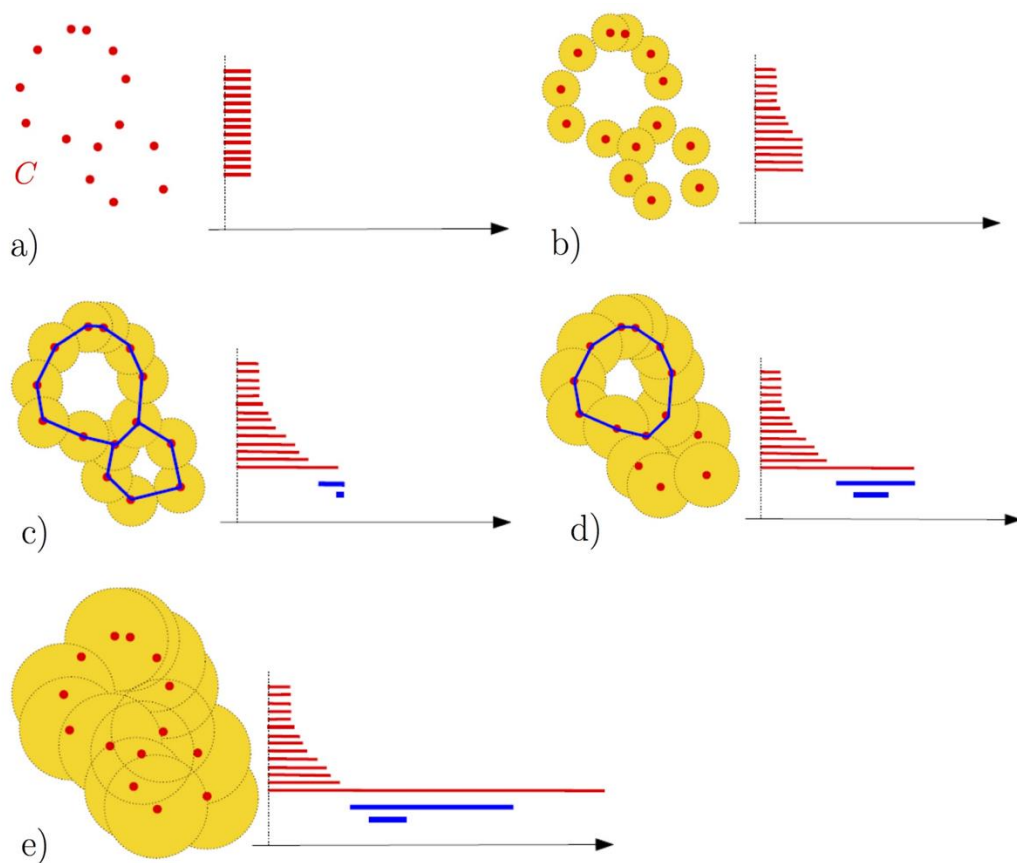
QTDA: [ncomms10138, 2005.02607, 2408.16934v3]

TDA: feature extraction, the “shape of data”, counts high dimensional holes [1710.04019]

Khovanov: [math/0201043]

Detects the unknot!

Unknotting is in NP and co-NP (like factoring).



QKhovanov: [2501.12378]

Easy invariants

Alexander polynomial is classically easy to compute as it has a free fermion interpretation.

$$\Delta_K(t) \doteq \det(A) = \int \left(\prod_{i=1}^n d\bar{\psi}_i d\psi_i \right) \exp \left(- \sum_{i,j=1}^n \bar{\psi}_i A_{ij} \psi_j \right)$$

But, Alexander polynomial enables quantum money [1004.5127]:

$$| \$_p \rangle = \frac{1}{\sqrt{N}} \sum_{G: A(G)=p} \sqrt{q(d(G))} |G\rangle$$

Barn-Natan's new Fun invariant [2509.18456] as a generalization of the Alexander polynomial.

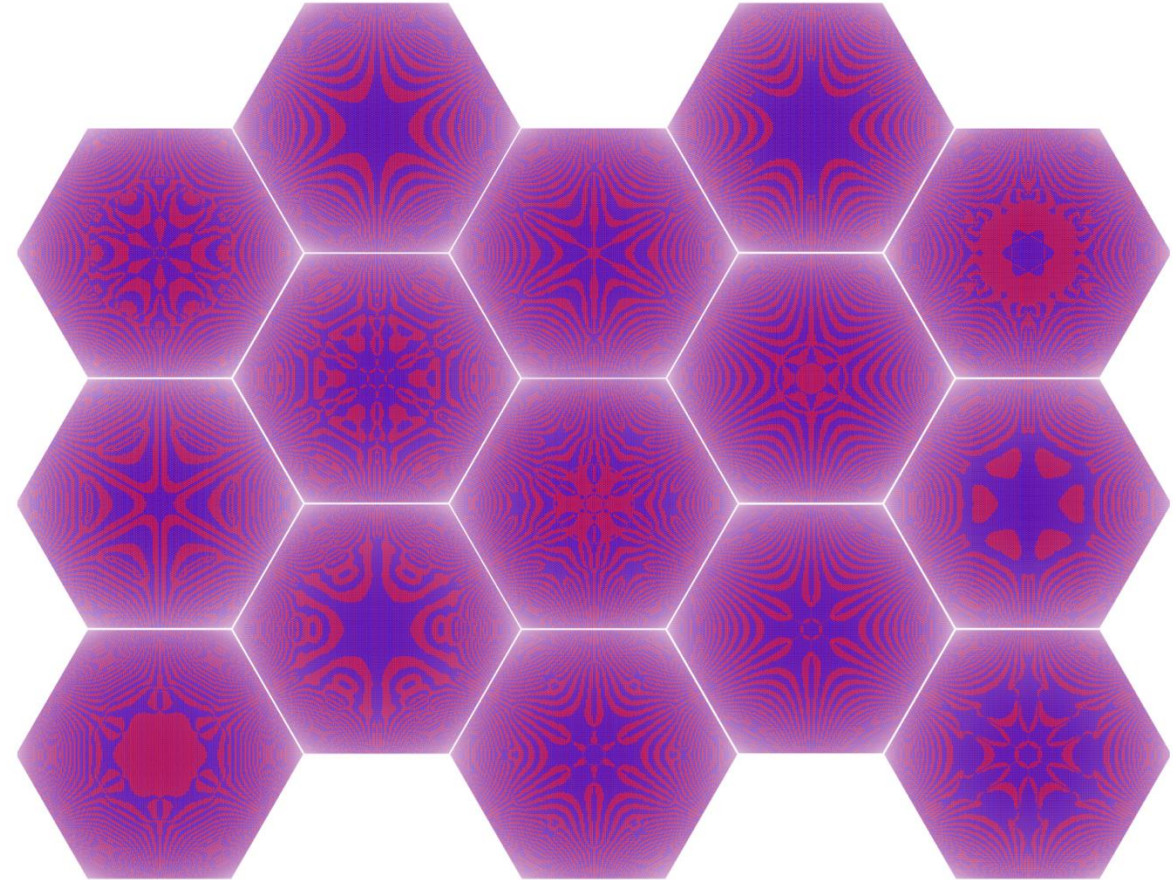
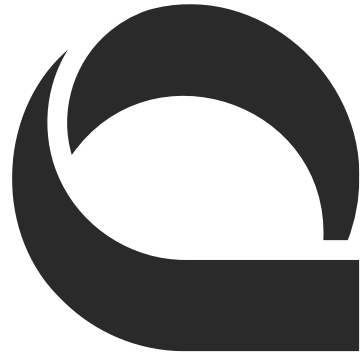


FIGURE 1.4. θ (hexagonal QR code only) of the 15 largest knots that we have computed by September 16, 2024. They are all “generic” in as much as we know, and they all have ≥ 300 crossings. The knots come from [DHOEBL]. Warning: Some screens/printers may introduce spurious Moiré interference patterns.



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